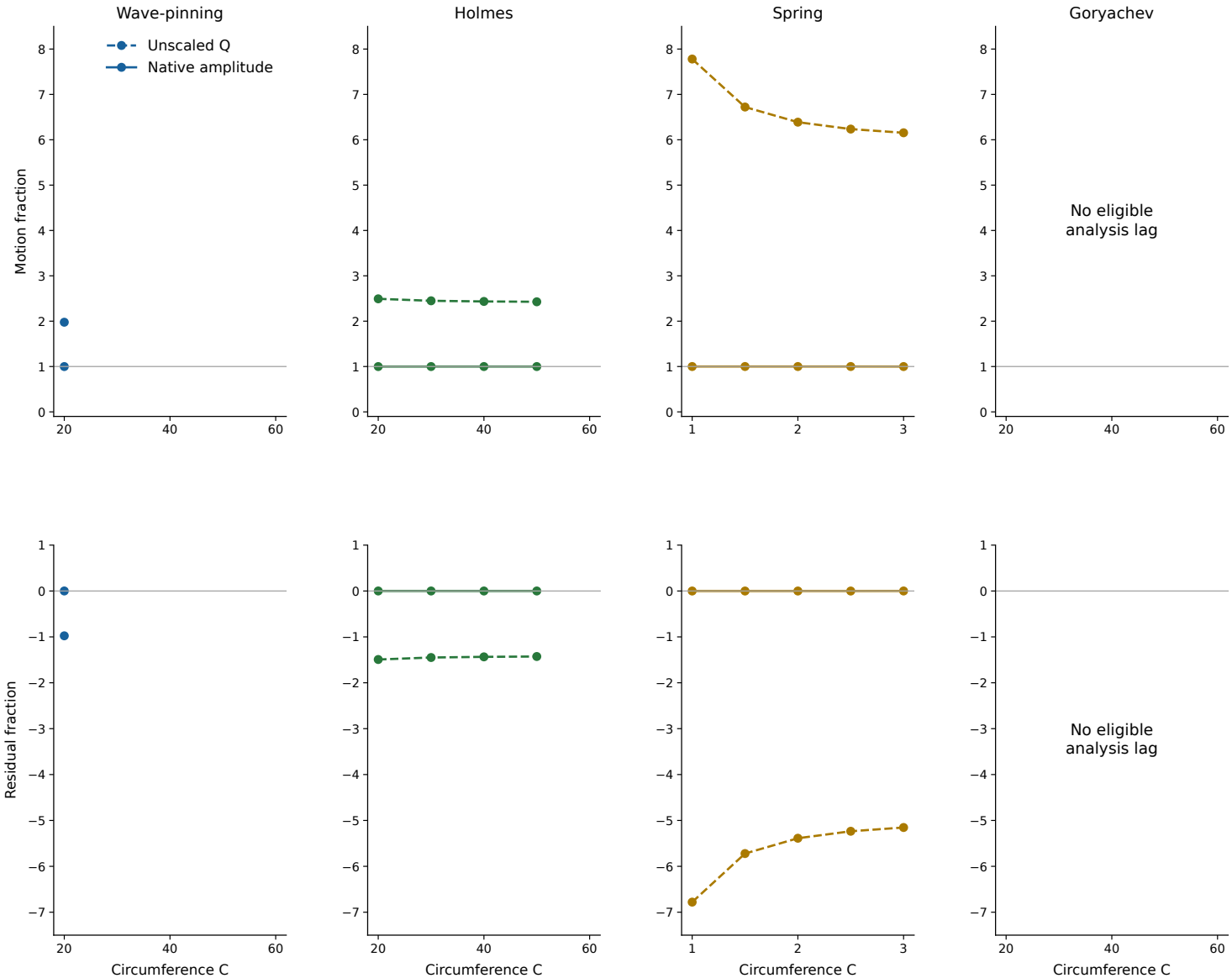
S4.A Native amplitude removes the point-region normalization artefact

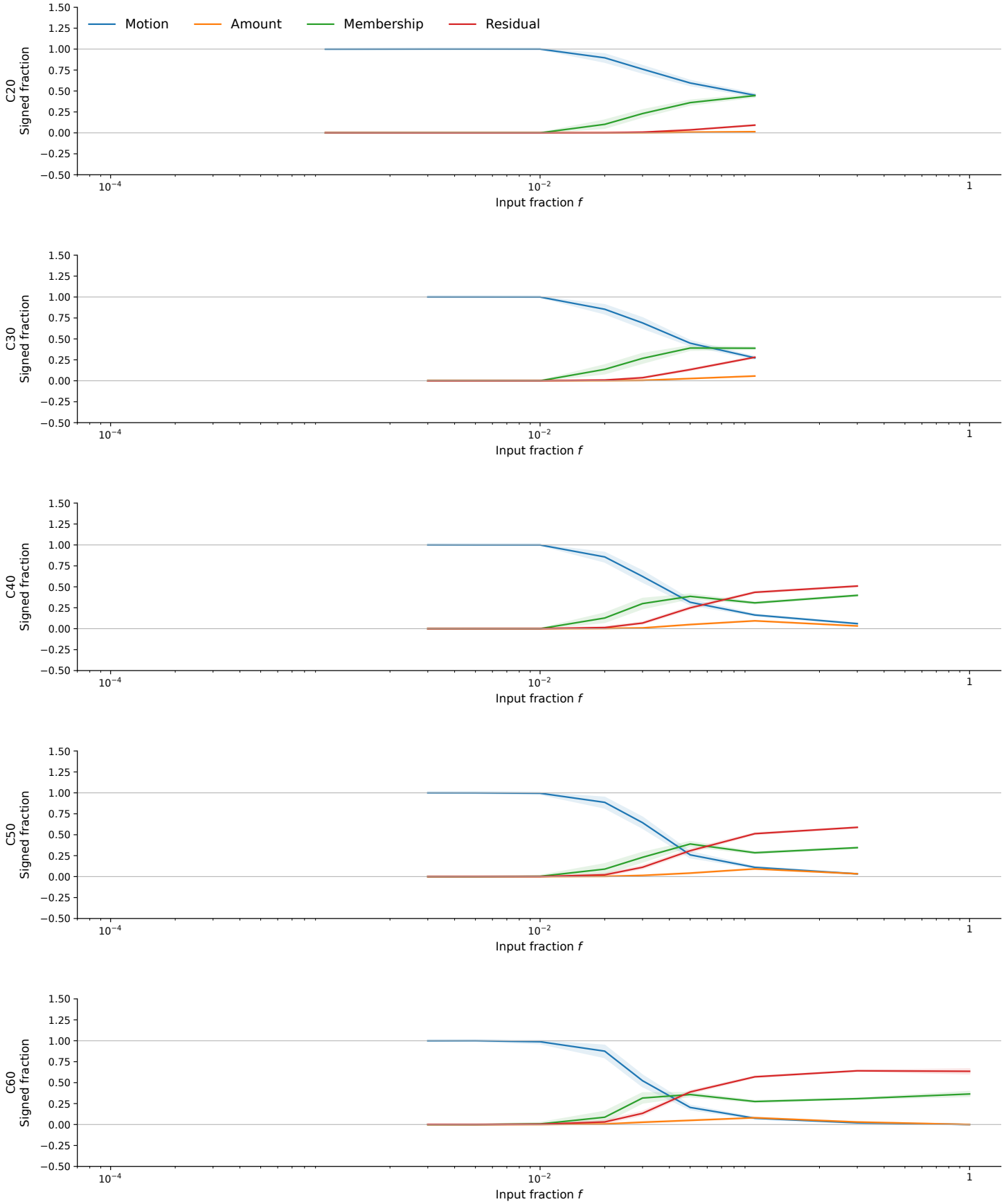


$$Q = \frac{\sum_i A_i e^{i\theta_i}}{\sum_i A_i}, \quad r_t = |P_t|, \quad \Delta P_j = \bar{r}\,\Delta Q_j \ (j = \mathrm{M}, \mathrm{A}, \mathrm{B}) \tag{S4.1}$$

$$\Delta P_{\mathrm{R}} = \Delta P - \bar{r}\,\Delta Q, \qquad \bar{r} = (r_0 + r_1)/2 \tag{S4.2}$$

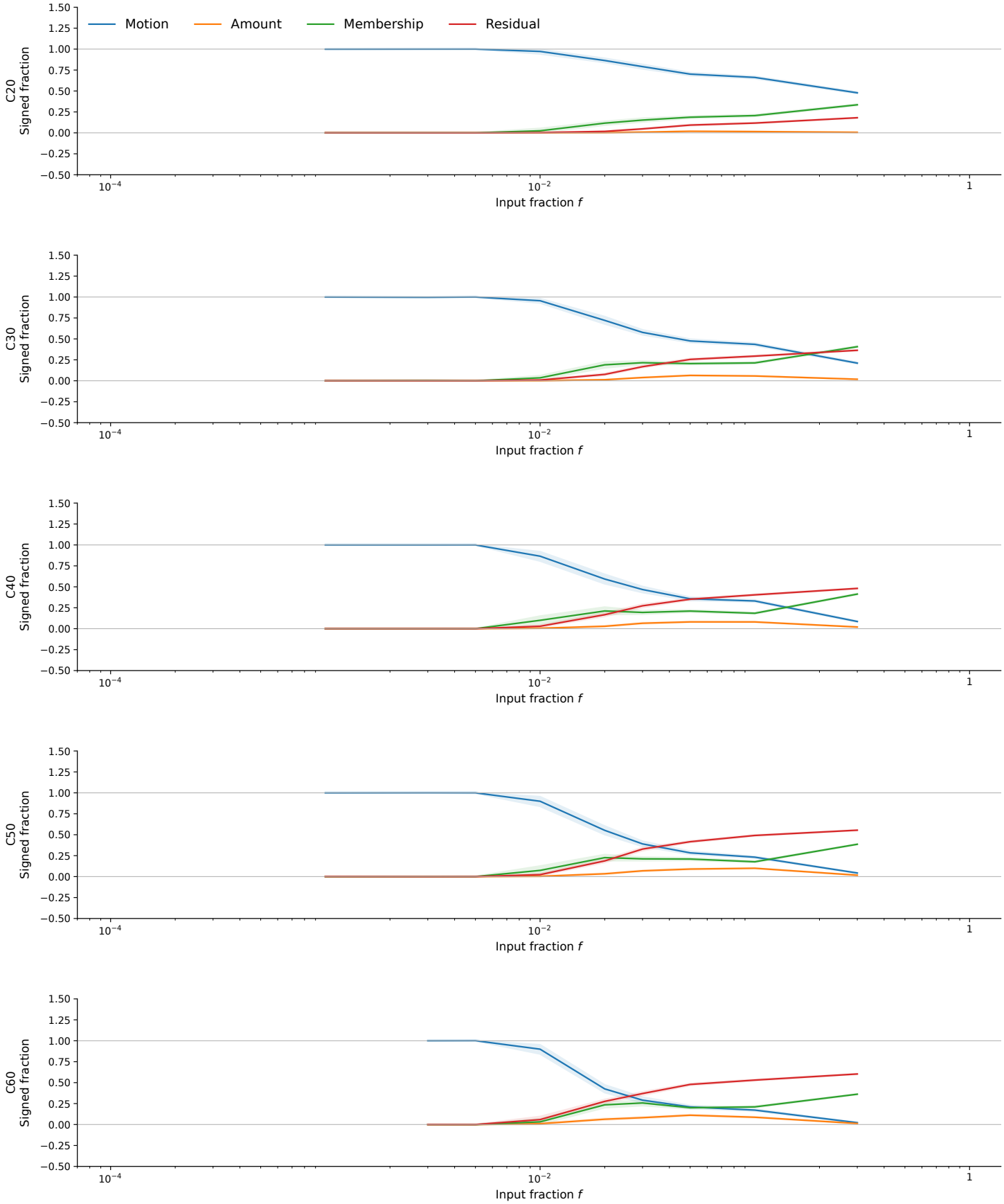
At f=0.001, the original unit-modulus single-region Q creates M≈1/|P| and R≈1−1/|P|.
The main allocation scales each Shapley ΔQ component by mean native amplitude. Pure rigid translation has M=1 and R=0 even with variable amplitude. The residual retains amplitude/shape/background effects. This remains an exact descriptive convention, not causal attribution. Missing Goryachev estimates remain absent.

S4.B Wave-pinning: amplitude-aware signed heading-loss allocation



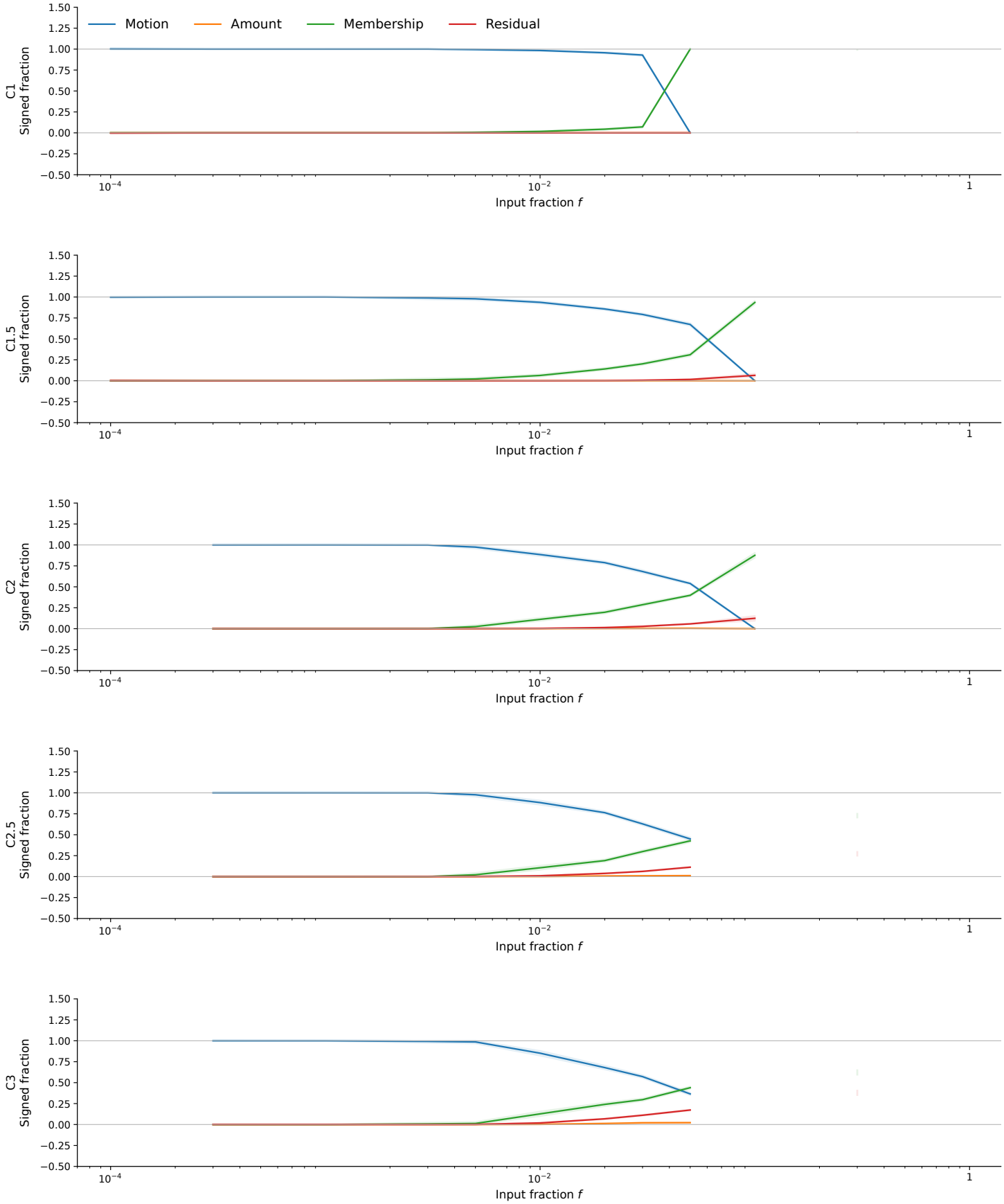
Pointwise whole-history bootstrap intervals (2,000 draws). Source CSV retains values outside the displayed limits.
Same endpoint pairs, lag and raw moments as the unscaled diagnostic; only the accounting convention changes.

S4.B Holmes: amplitude-aware signed heading-loss allocation



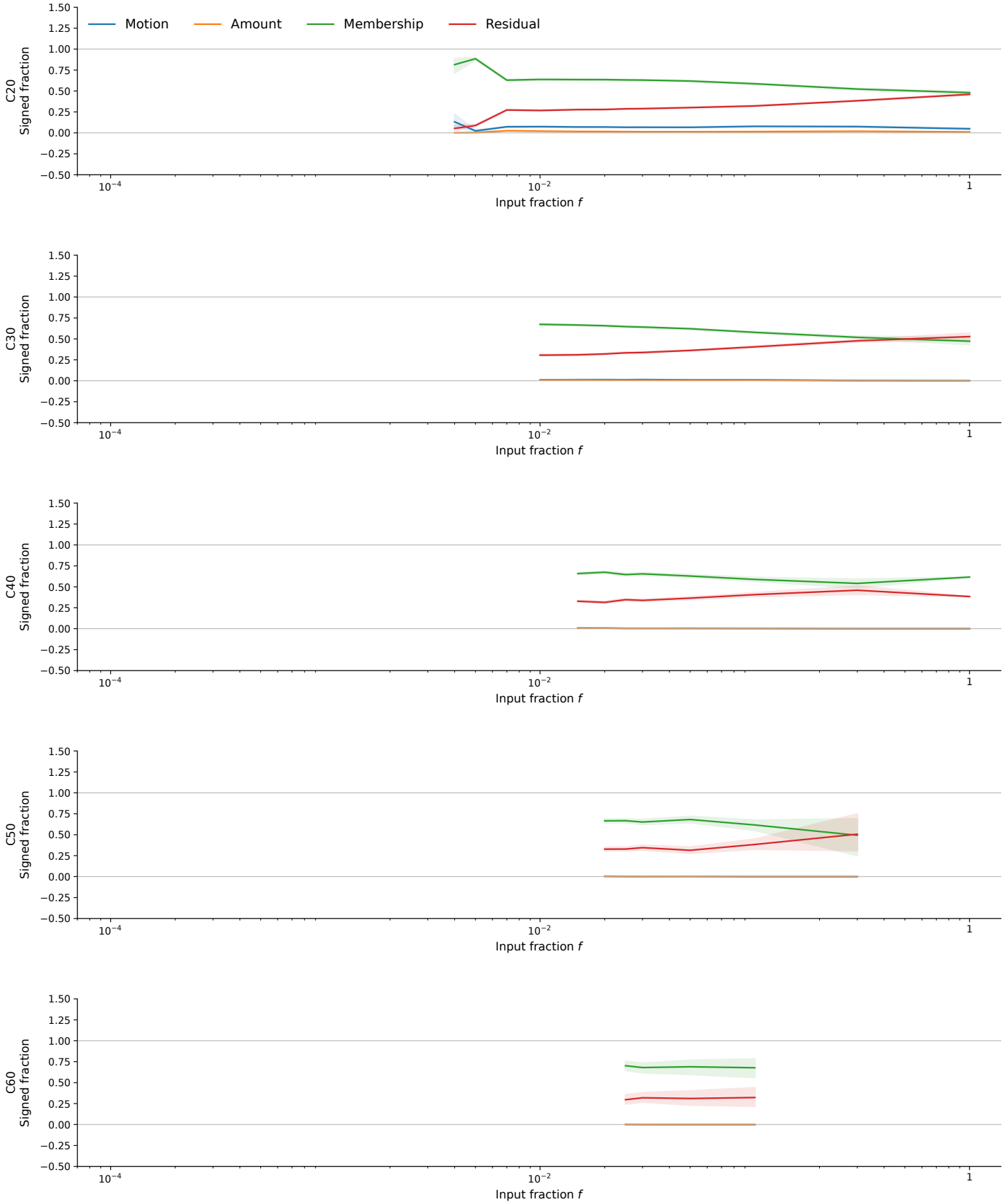
Pointwise whole-history bootstrap intervals (2,000 draws). Source CSV retains values outside the displayed limits.
Same endpoint pairs, lag and raw moments as the unscaled diagnostic; only the accounting convention changes.

S4.B Spring: amplitude-aware signed heading-loss allocation



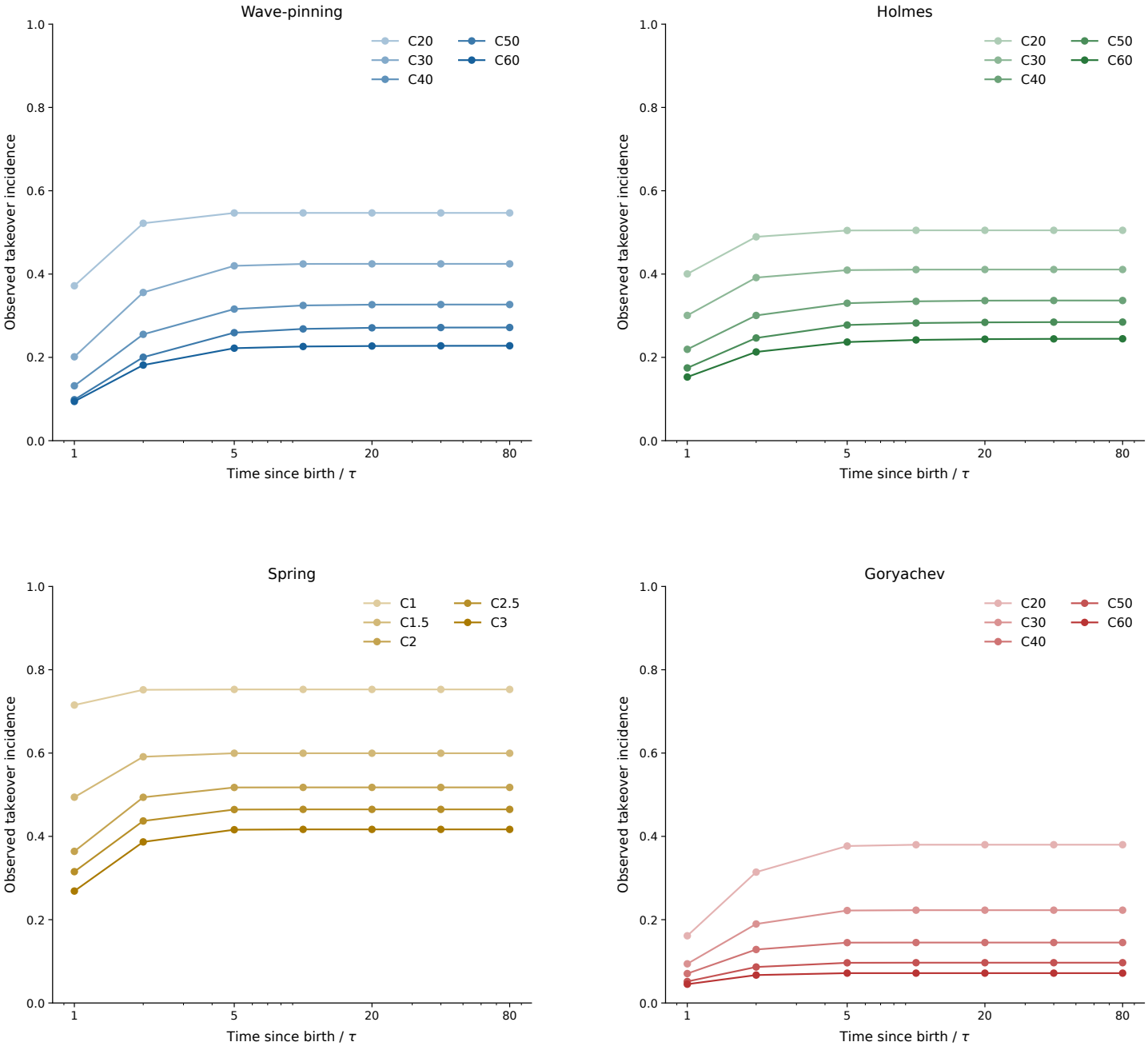
Pointwise whole-history bootstrap intervals (2,000 draws). Source CSV retains values outside the displayed limits.
Same endpoint pairs, lag and raw moments as the unscaled diagnostic; only the accounting convention changes.

S4.B Goryachev: amplitude-aware signed heading-loss allocation



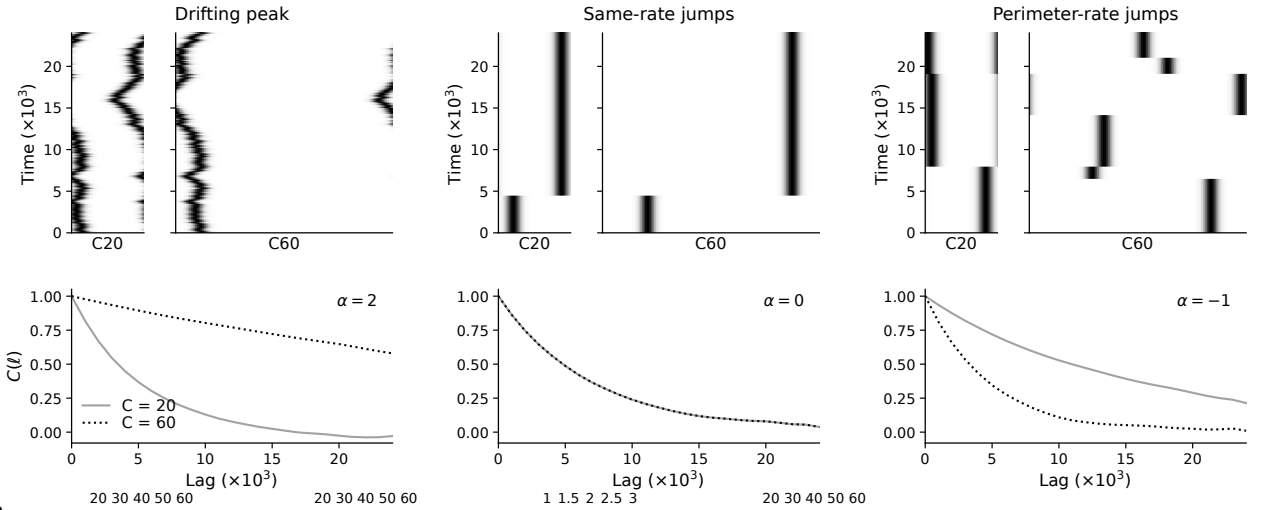
Pointwise whole-history bootstrap intervals (2,000 draws). Source CSV retains values outside the displayed limits.
Same endpoint pairs, lag and raw moments as the unscaled diagnostic; only the accounting convention changes.

S4.C A fixed follow-up cohort separates observation window from outcome

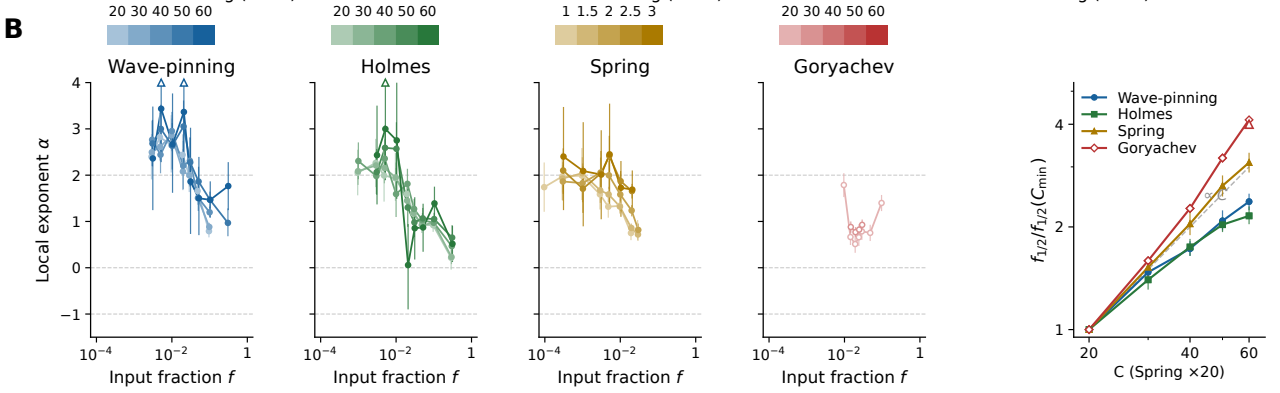


Certified births with at least 80τ available record follow-up; the same cohort is used at every plotted horizon.
All distance bins are included, and f>0.01 is pooled. Curves are descriptive; no extrapolation to lifetime success.
A lower incidence at 20τ can reflect delayed takeover, a lower eventual incidence, or both.

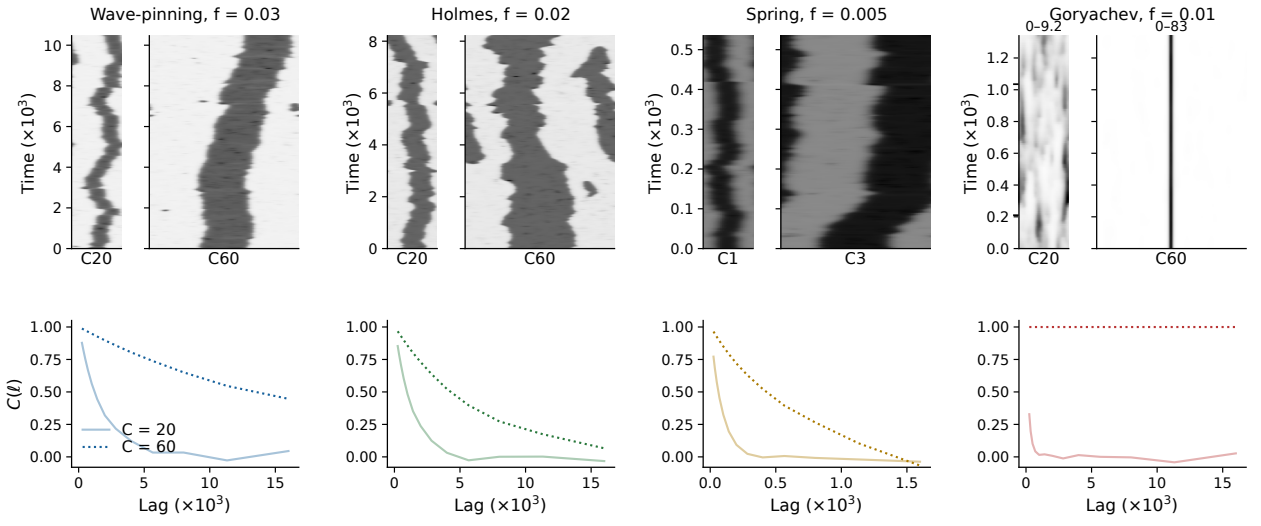
A


Drifting peak
Same-rate jumps
Perimeter-rate jumps
Time (×10³)
C20
C60
α = 2
α = 0
α = −1
C(ℓ)
C = 20
C = 60
Lag (×10³)

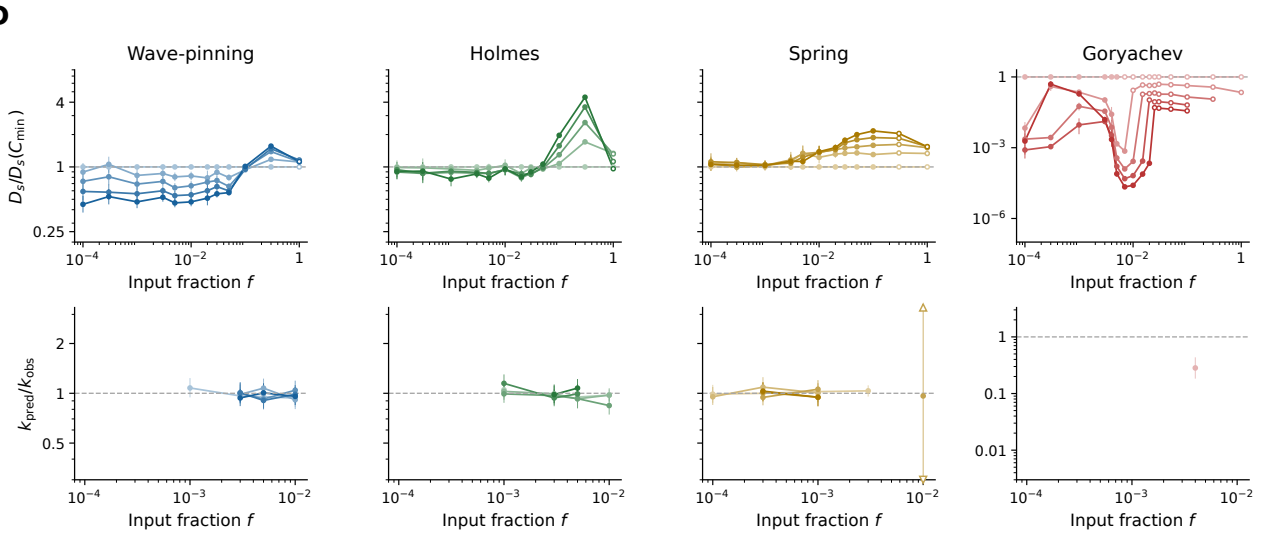
B


20 30 40 50 60
1 1.5 2 2.5 3
Wave-pinning
Holmes
Spring
Goryachev
Local exponent α
Input fraction f
Wave-pinning
Holmes
Spring
Goryachev
f1/2/f1/2(Cmin)
C (Spring ×20)

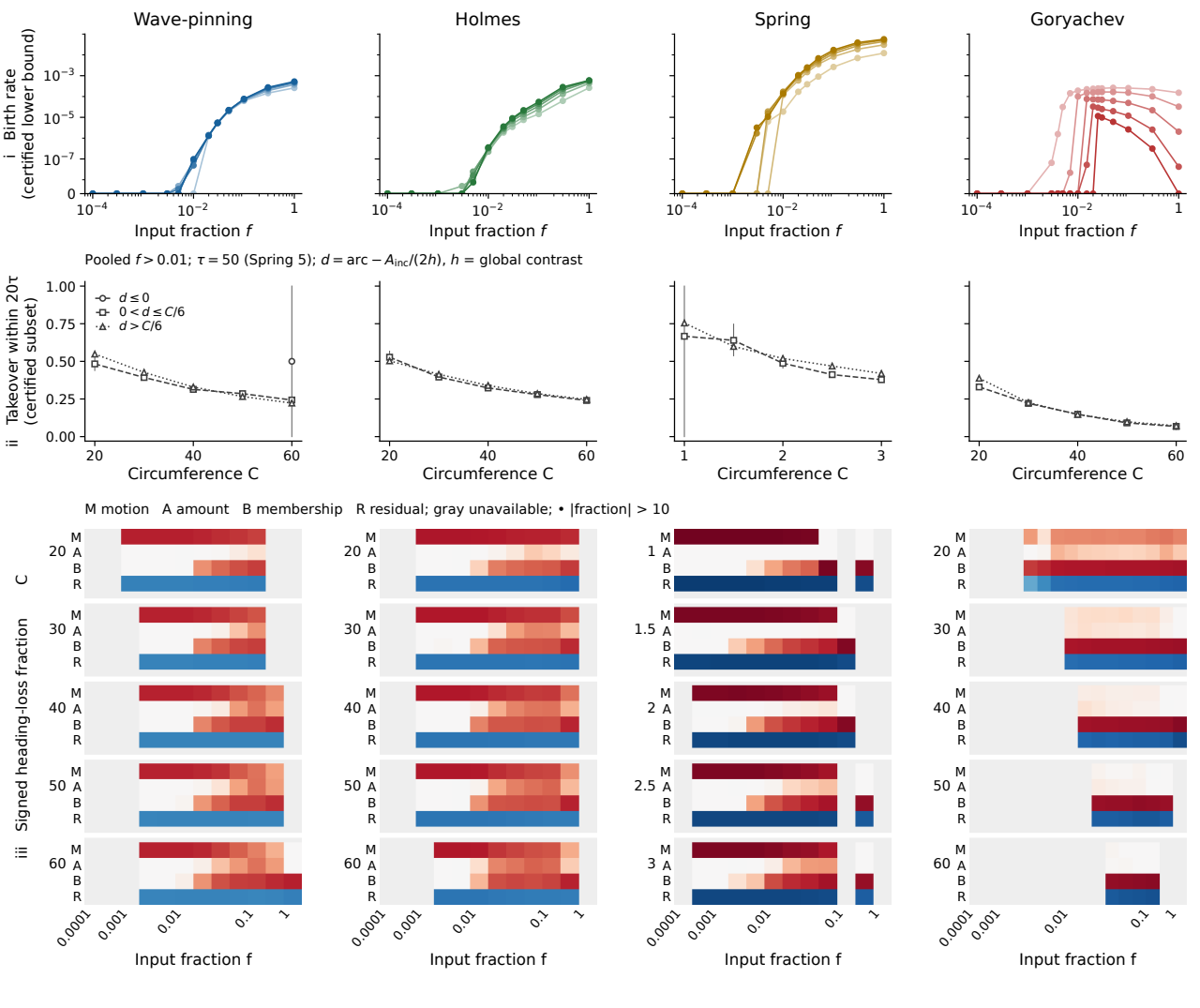
C


Wave-pinning, f = 0.03
Holmes, f = 0.02
Spring, f = 0.005
Goryachev, f = 0.01
0–9.2
0–83
Time (×10³)
C20
C60
C1
C3
C(ℓ)
C = 20
C = 60
Lag (×10³)

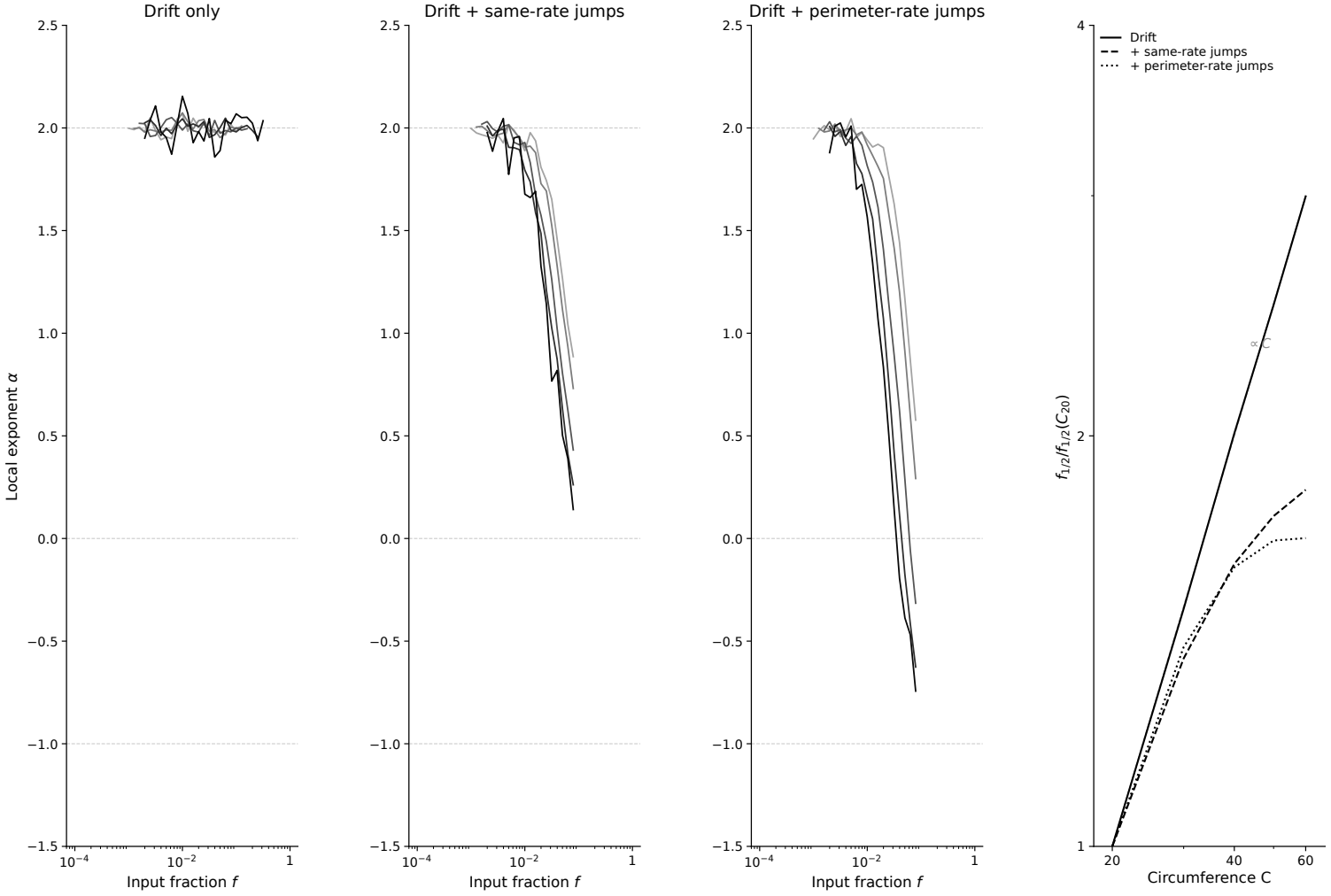
D


Wave-pinning
Holmes
Spring
Goryachev
Ds/Ds(Cmin)
kpred/kobs
Input fraction f

E

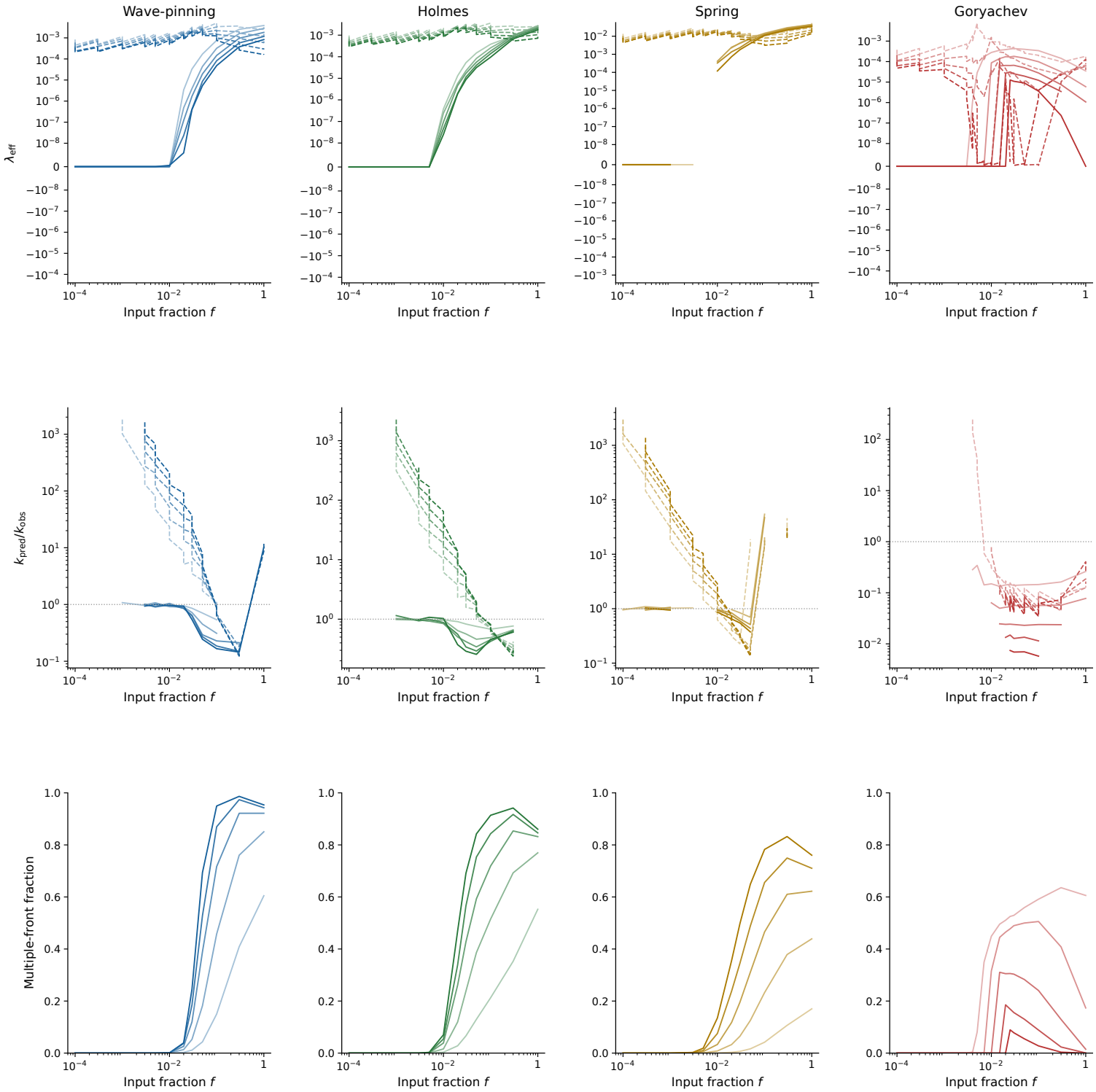

Wave-pinning
Holmes
Spring
Goryachev
i Birth rate (certified lower bound)
Input fraction f
Pooled f > 0.01; τ = 50 (Spring 5); d = arc − Ainc/(2h), h = global contrast
ii Takeover within 20τ (certified subset)
d ≤ 0
0 < d ≤ C/6
d > C/6
Circumference C
M motion A amount B membership R residual; gray unavailable; • |fraction| > 10
iii Signed heading-loss fraction
C
Input fraction f

S4.1 Displaced null-mixture exponents and tolerance folds



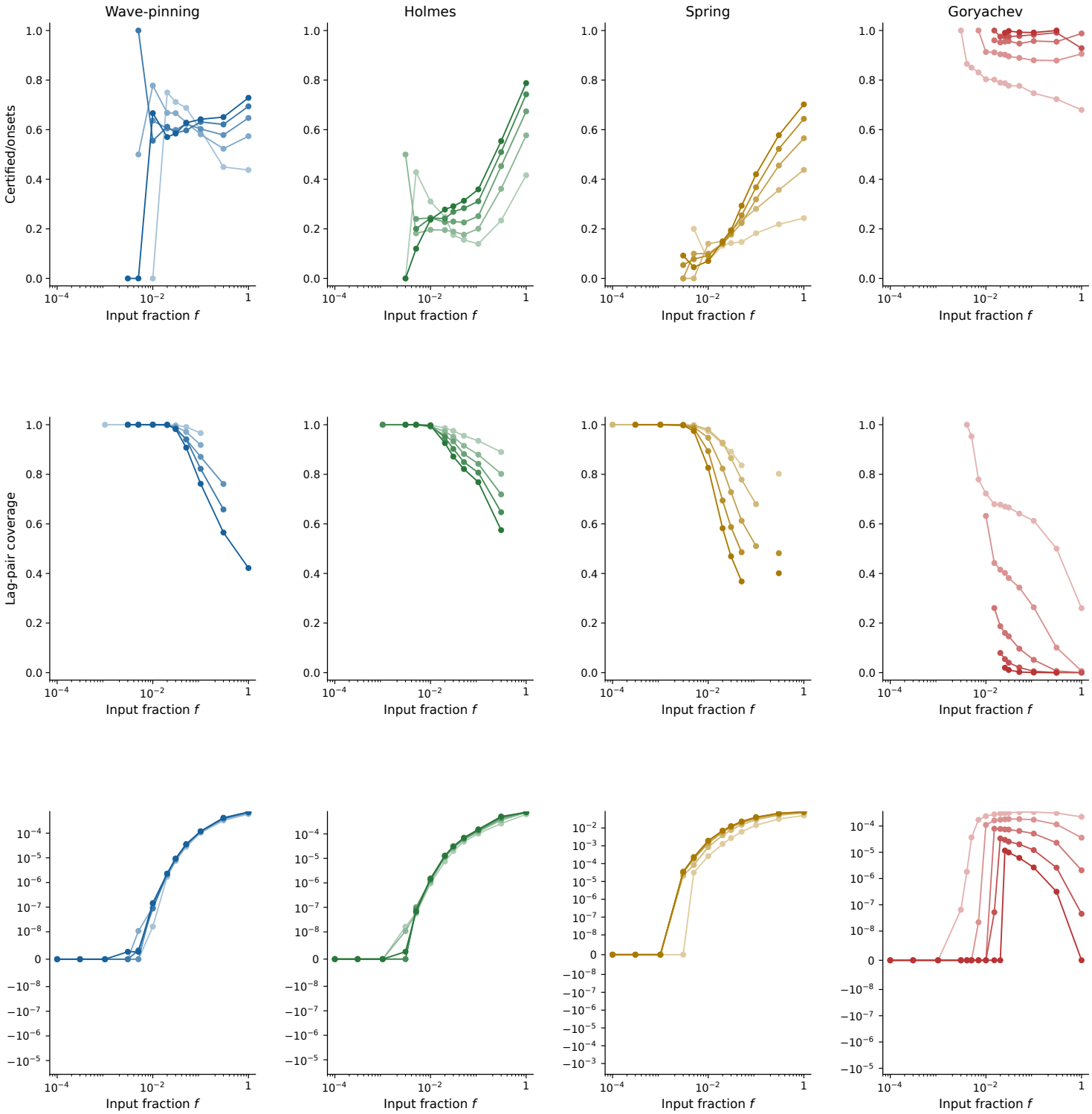
Historical simulated nulls; unchanged source records. These diagnostic curves were removed from the main figure.

S4.2 Displaced drift/relocation diagnostics



Solid: tracked; dashed: detector-free fit. All inputs and failures retained, with no rare-multiple filter. Native clocks differ across models.

S4.3 Identification and endpoint coverage



Certified births are lower bounds. All persistent secondary onsets give upper bounds including splits. Different certification fractions prevent population-level comparisons of takeover probability.

S4.4 Missing information and exact scope of the birth observable

Requested split counts are unavailable for every one of the 260 conditions.

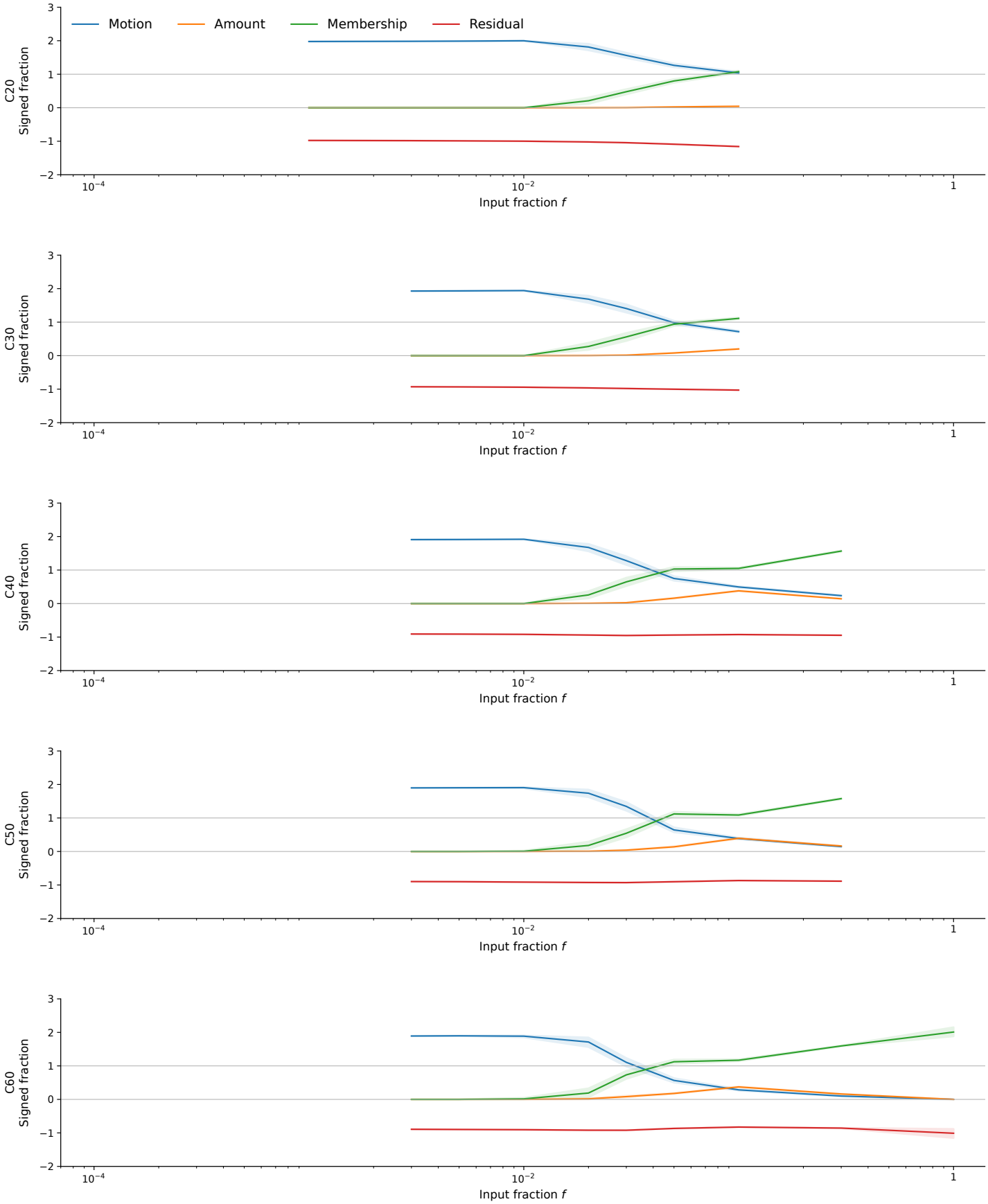
The compact producer stored maximum-overlap identities, excess-weighted contour centroids, full-basin amounts and contour-cell widths. It discarded contour masks/endpoints, parent IDs, per-region plateau heights and event lists. A centroid and width do not determine the component interval, because the centroid is weighted by the unknown within-component profile.

A certified no-parent birth is a new identity that persists for at least five frames and whose conservative angular support enclosure misses every previous enclosure. This is a subset of all no-overlap births. No uncertified track is called a split or a true birth.

Incumbent half-width is estimated as full-basin amount / (2 × current global contrast), using the prior frame. It is an explicit plateau approximation, not a recovered local plateau. Takeovers require five consecutive raw dominant-ID frames, within a common follow-up of 20 noise blocks (1,000 chemical units; 100 Spring). Shorter follow-up is excluded. Distances are grouped as $d \le 0$, $0 < d \le C/6$ and $d > C/6$; designed inputs $f > 0.01$ are pooled.

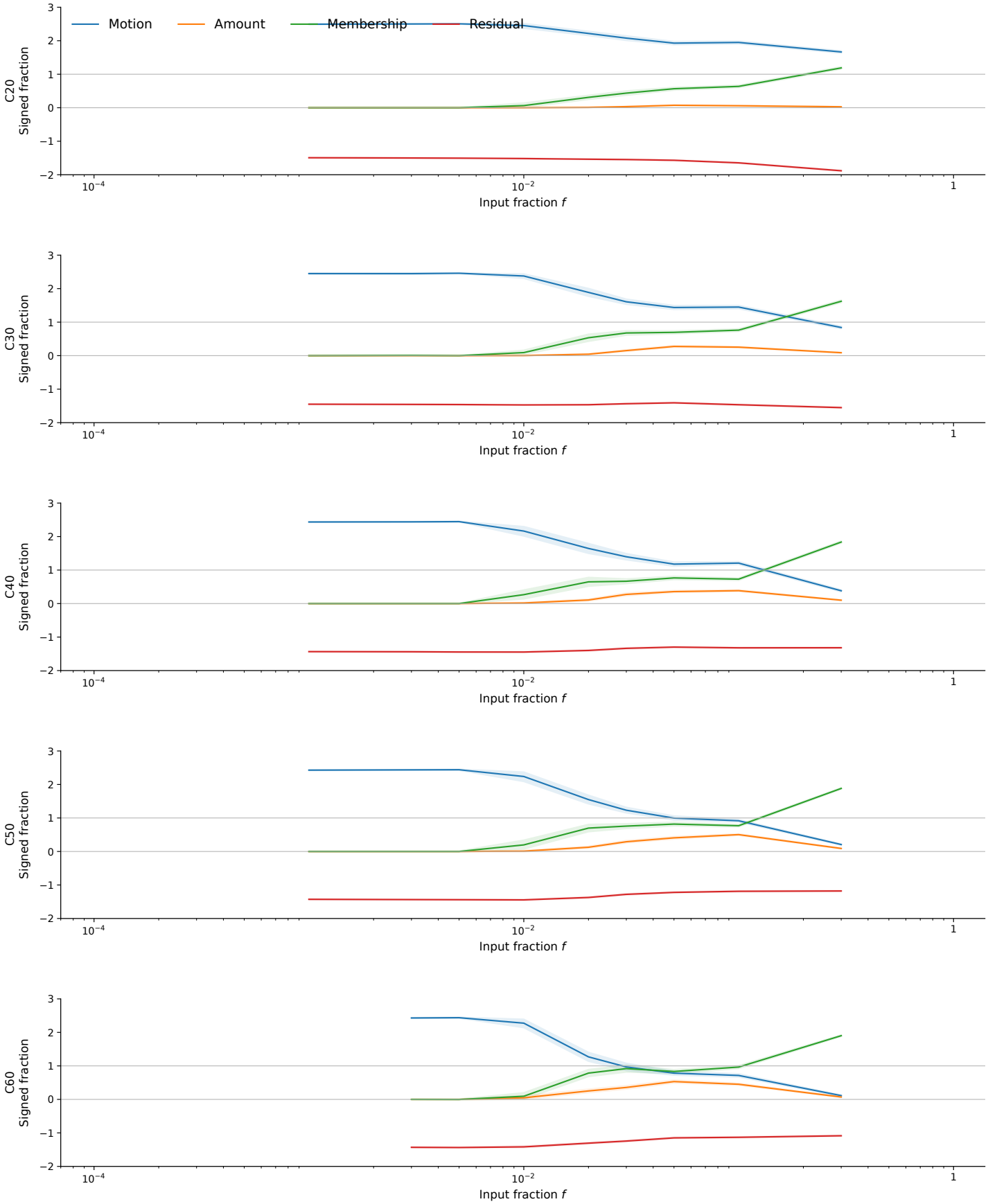
The historical aggregate region audits used a two-frame pairwise-dominance rule and include split onsets. They cannot replace the new five-frame event definition. No noise runs were added.

S4.5 Wave-pinning: signed heading-loss fractions



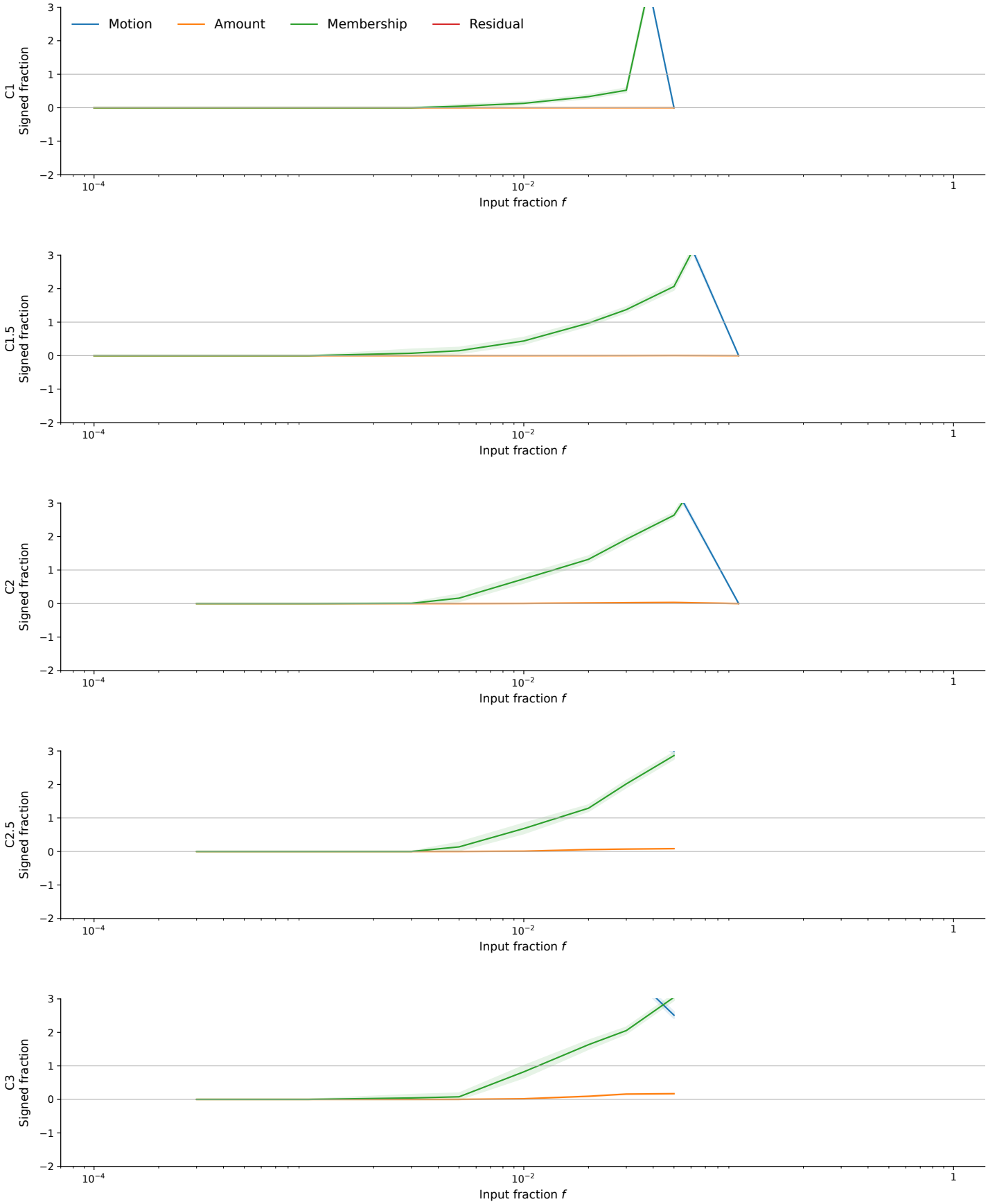
Pointwise whole-seed bootstrap intervals (2,000 draws). Values outside displayed limits remain in CSV. Fractions sum to one; negative values represent cancellation. This allocation is descriptive and convention dependent.

S4.5 Holmes: signed heading-loss fractions



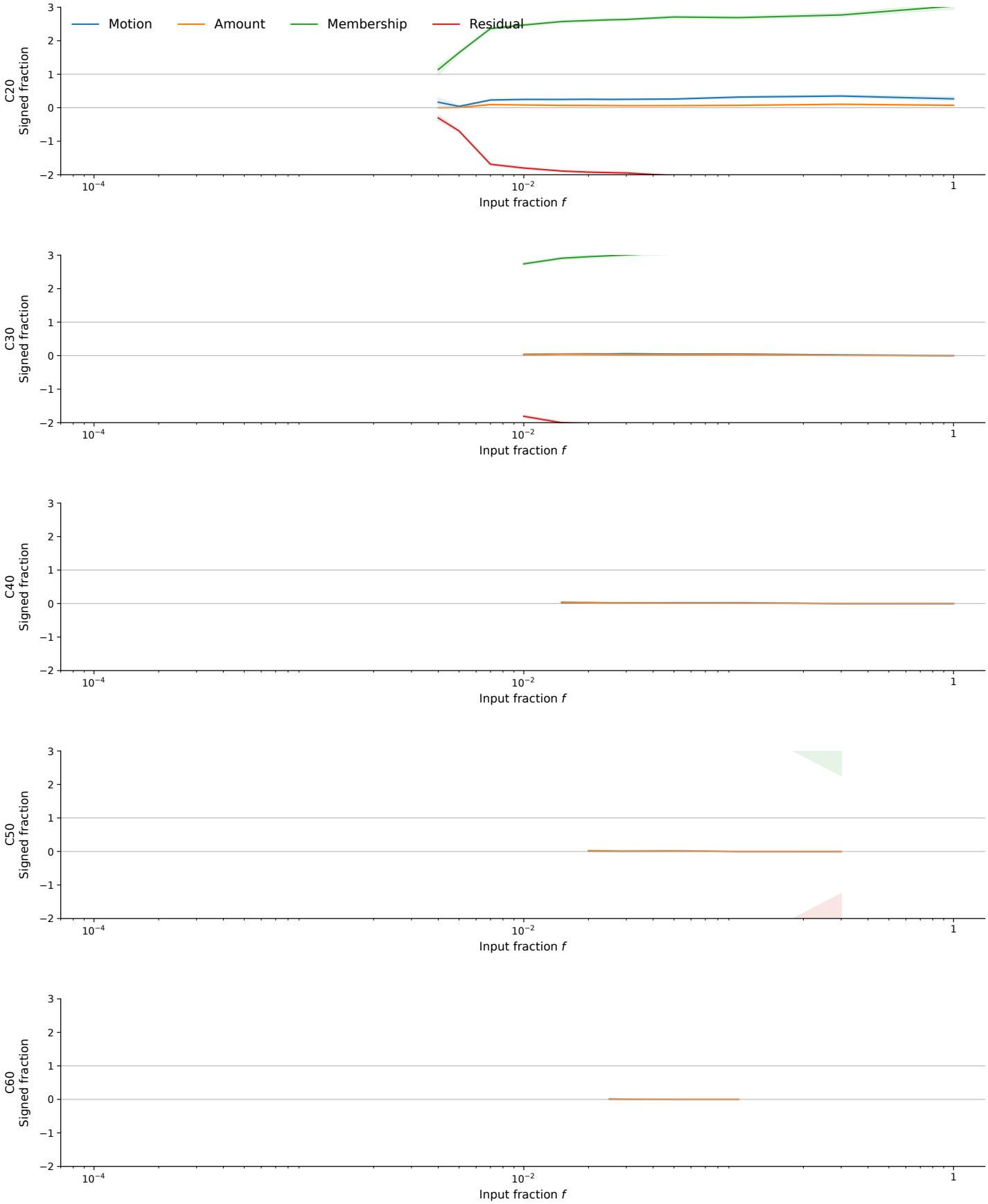
Pointwise whole-seed bootstrap intervals (2,000 draws). Values outside displayed limits remain in CSV. Fractions sum to one; negative values represent cancellation. This allocation is descriptive and convention dependent.

S4.5 Spring: signed heading-loss fractions

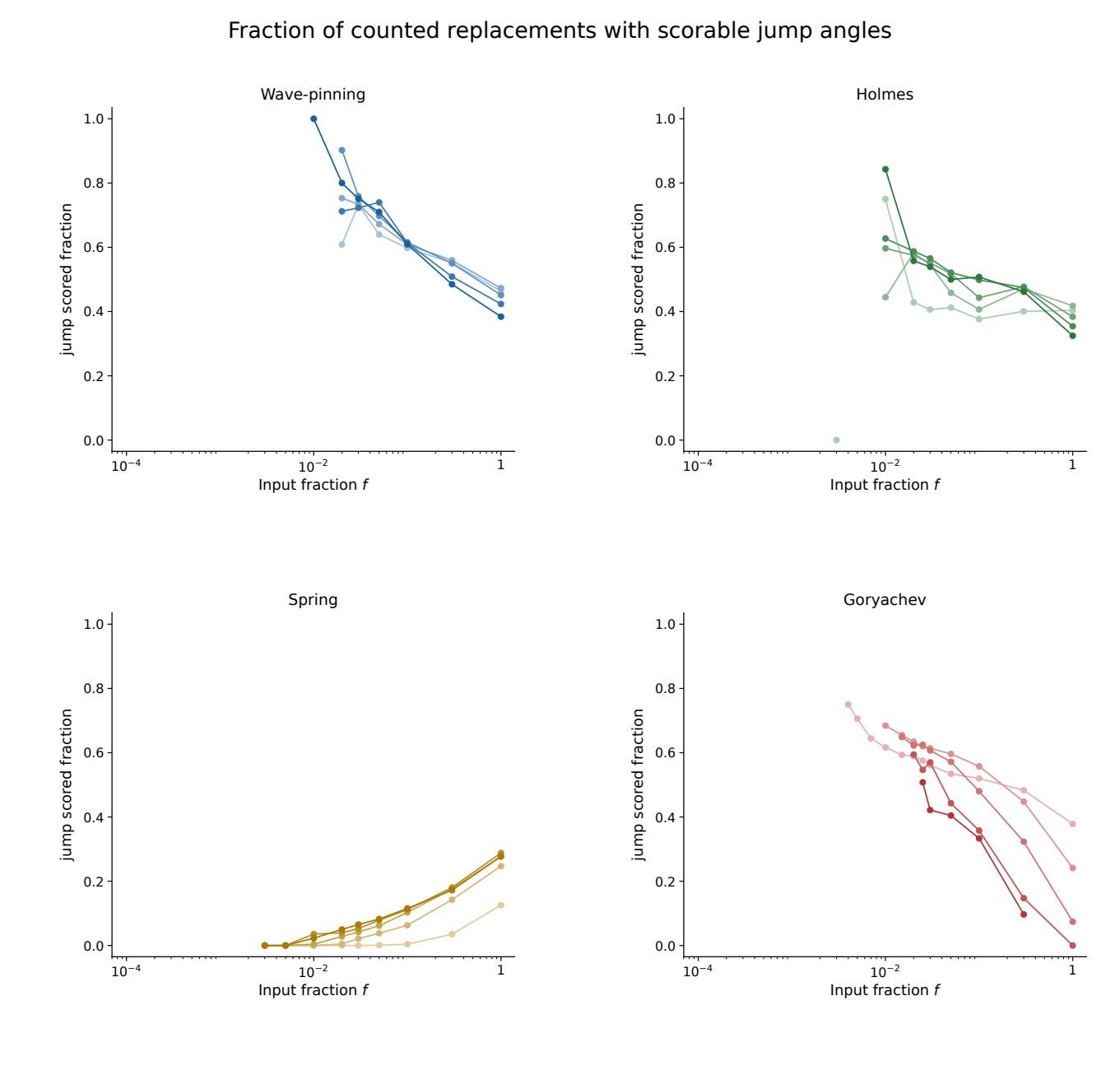


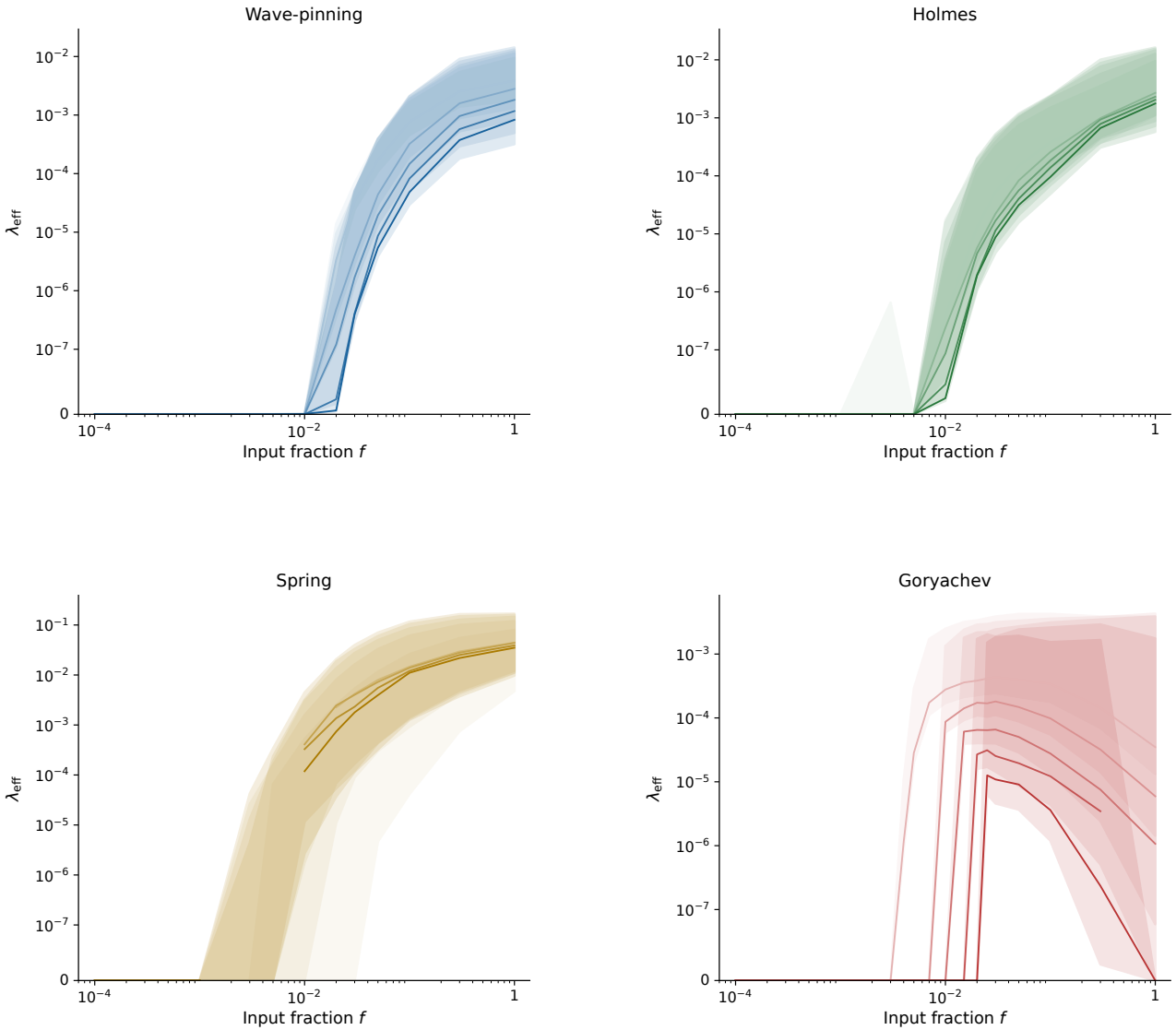
Pointwise whole-seed bootstrap intervals (2,000 draws). Values outside displayed limits remain in CSV. Fractions sum to one; negative values represent cancellation. This allocation is descriptive and convention dependent.

S4.5 Goryachev: signed heading-loss fractions



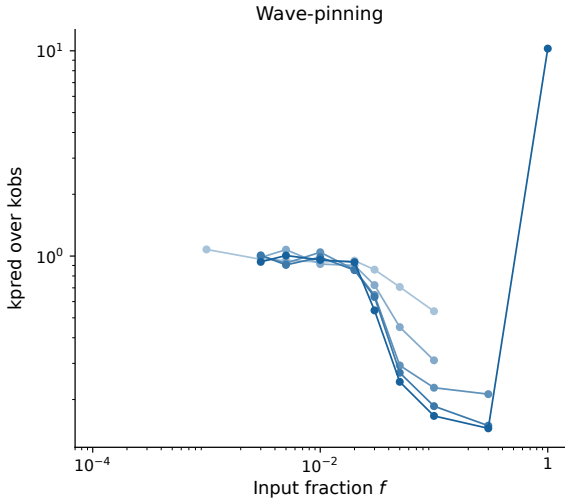
Pointwise whole-seed bootstrap intervals (2,000 draws). Values outside displayed limits remain in CSV. Fractions sum to one; negative values represent cancellation. This allocation is descriptive and convention dependent.

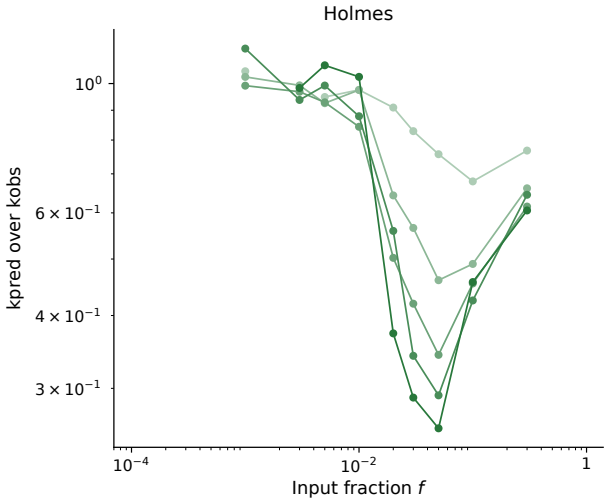

Fraction of counted replacements with scorable jump angles
Wave-pinning
Holmes
Spring
Goryachev
jump scored fraction
Input fraction f
1.0
0.8
0.6
0.4
0.2
0.0
10^{-4}
10^{-2}
1

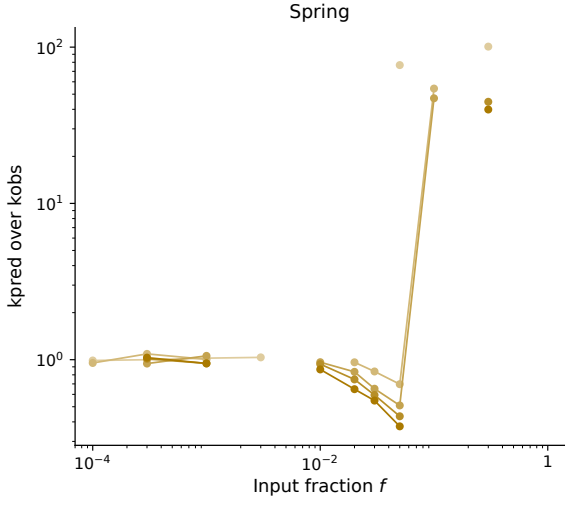


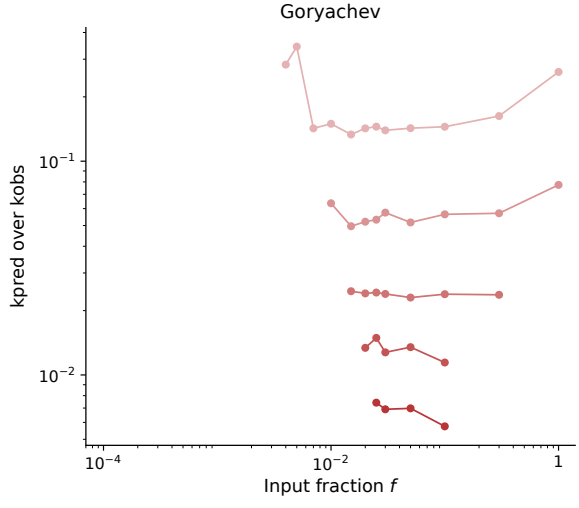
Shading: every unscored event assigned loss 0 or 2. Lines: all-event rate × mean loss among scorable events. These identification bounds are distinct from bootstrap intervals.

Scorable events: full predicted / observed rate range

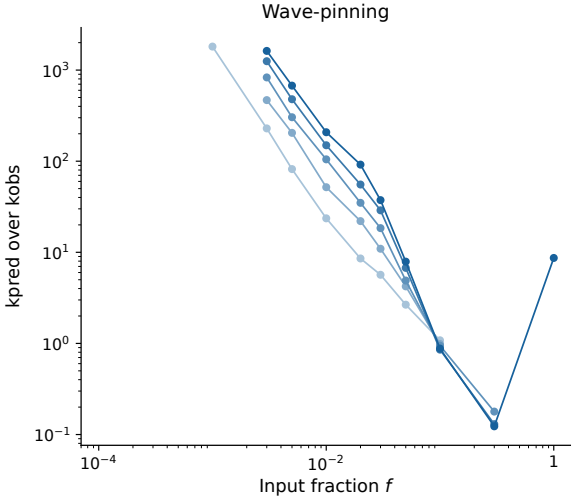


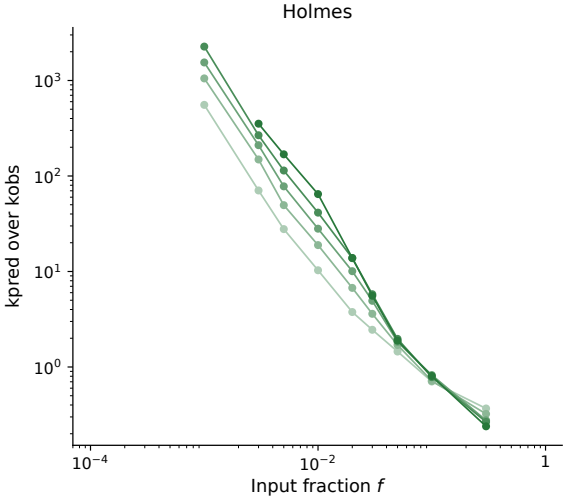


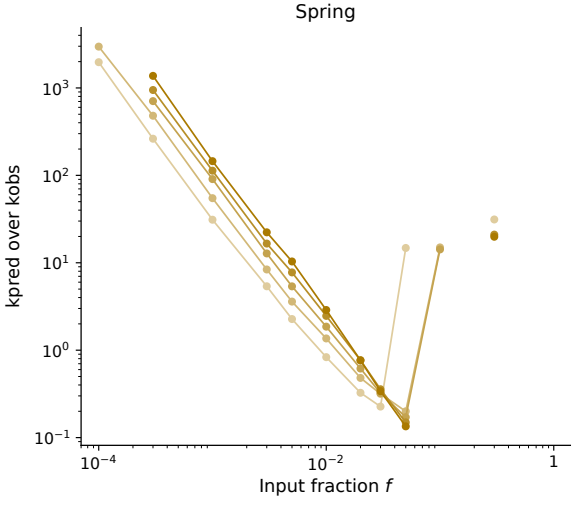


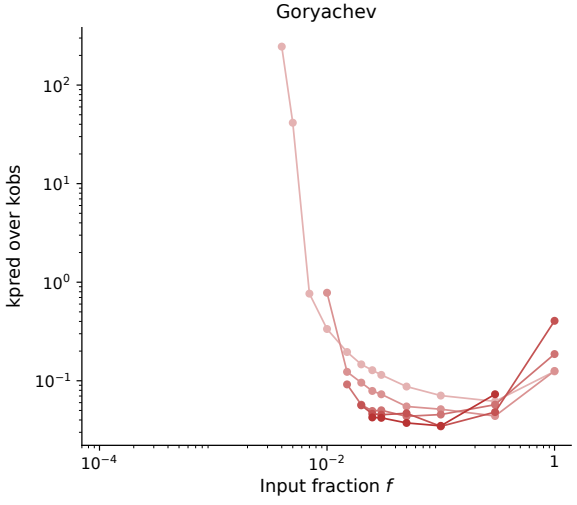


Mixture at shortest cadence: full predicted / observed rate range

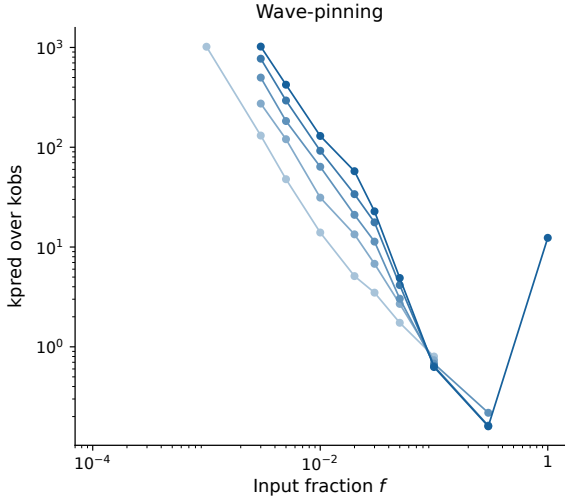


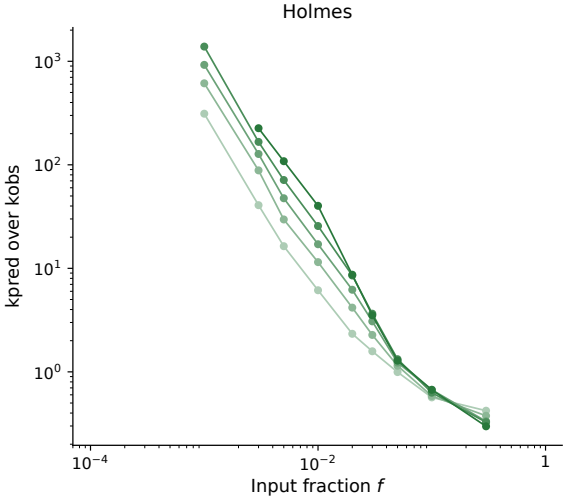


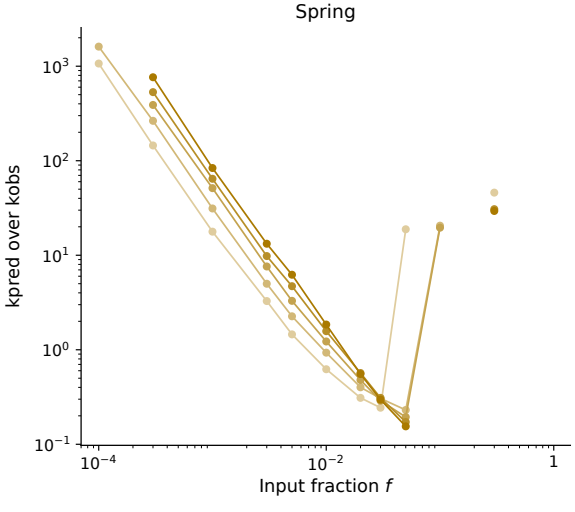


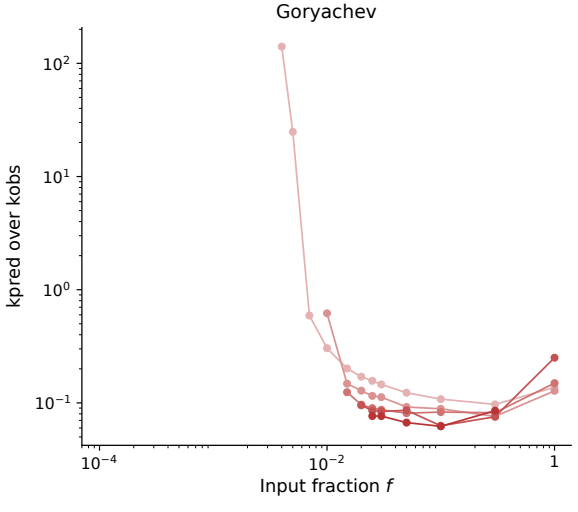


Mixture at twice cadence: full predicted / observed rate range

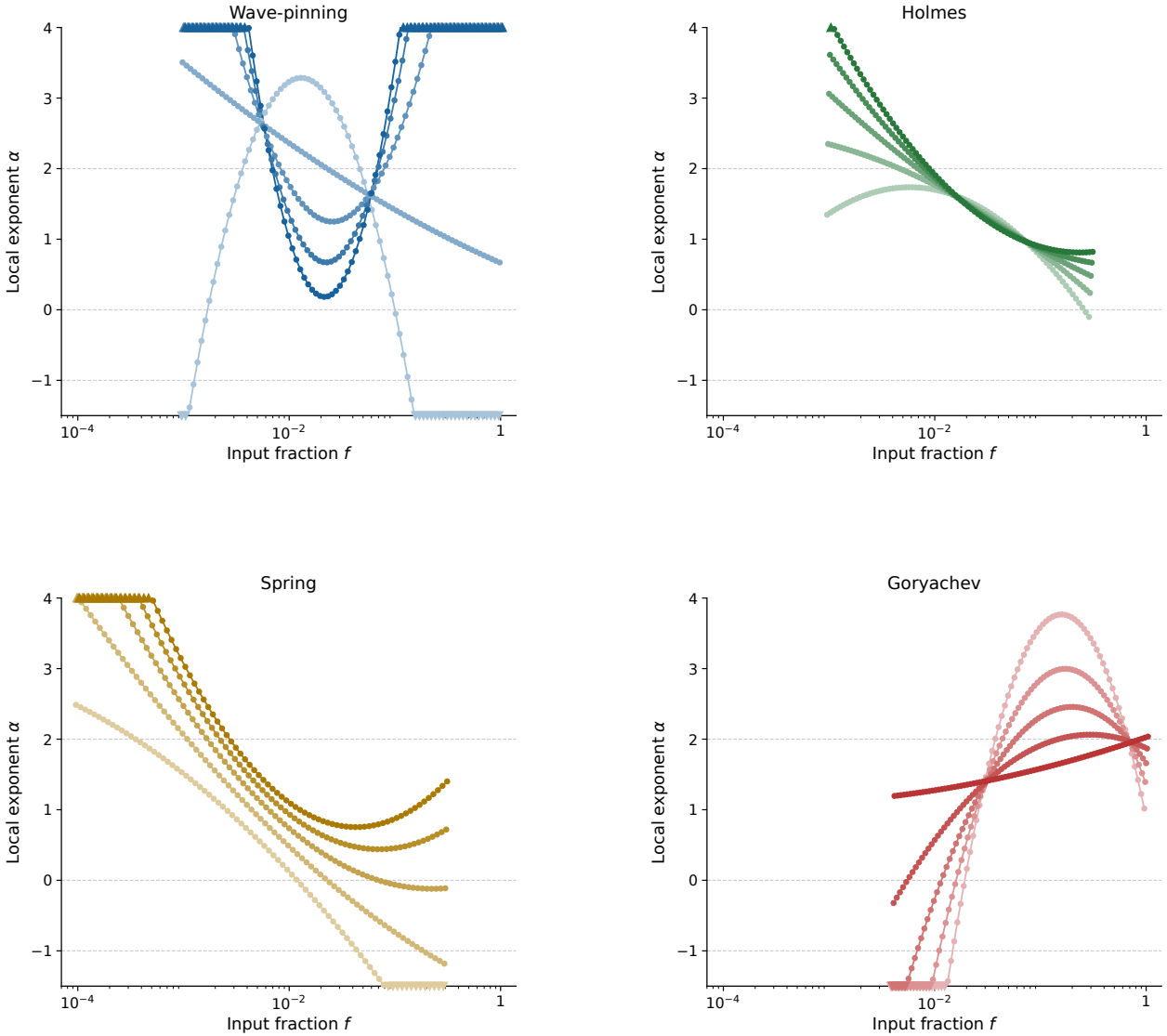




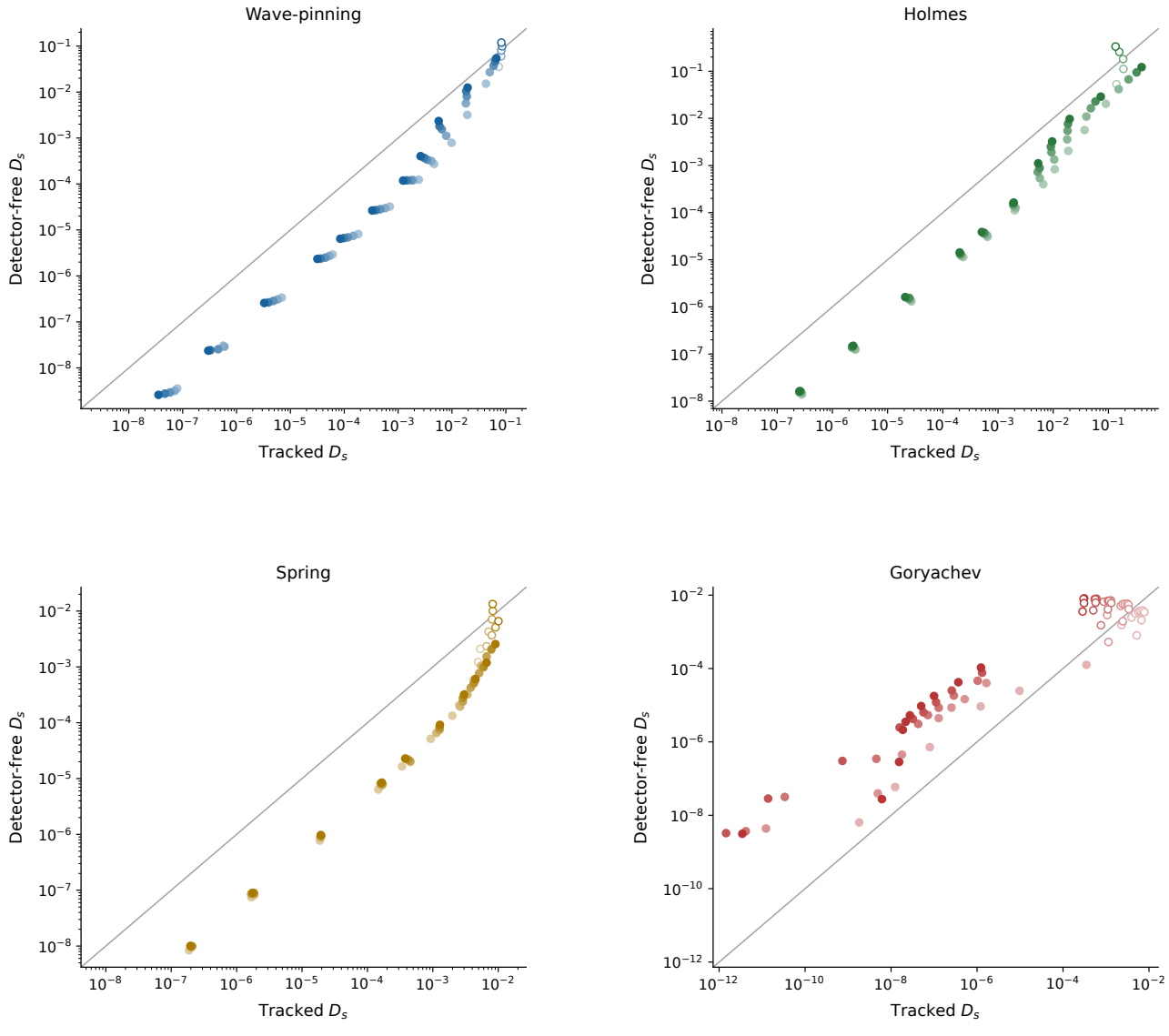




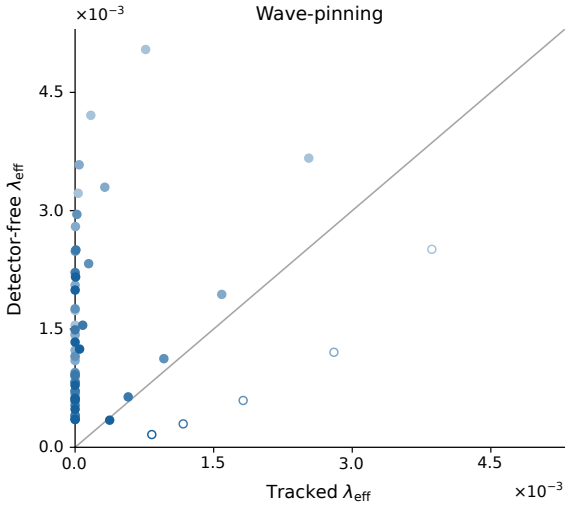
Smooth diagnostic size–input surface

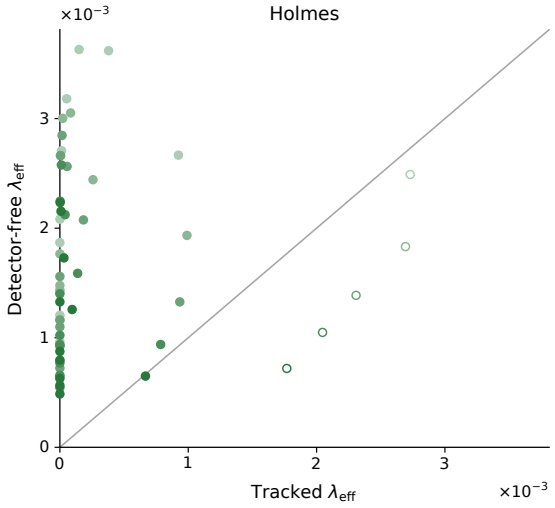


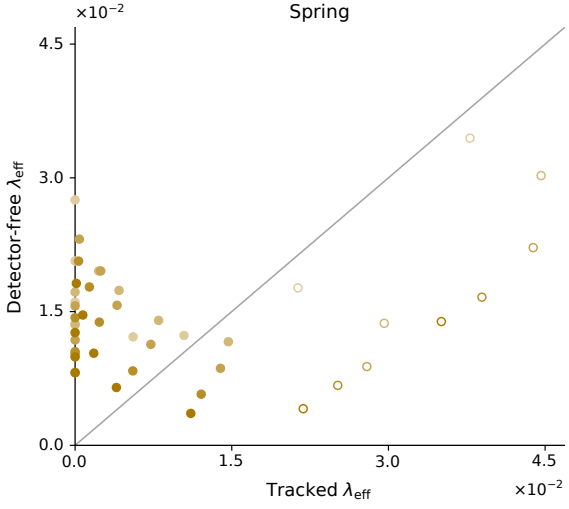
Fixed degree-two tensor polynomial in ln k versus ln circumference and ln input; diagnostic only. No missing local estimate is filled in the main figure.

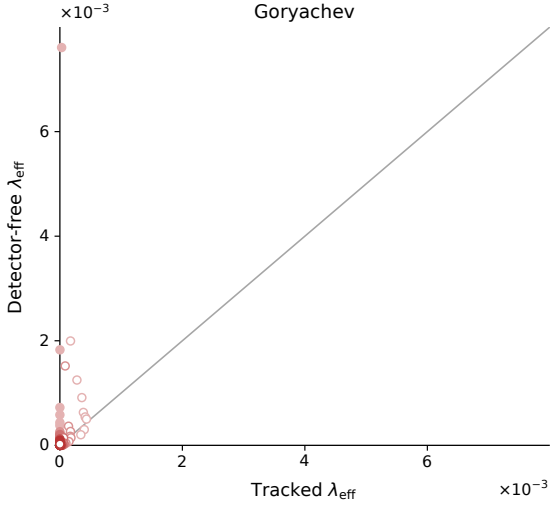

Detector-free versus tracked D_s; five circumference shades
Wave-pinning
Holmes
Spring
Goryachev
Detector-free D_s
Tracked D_s

Detector-free versus tracked λ_{eff}; five circumference shades

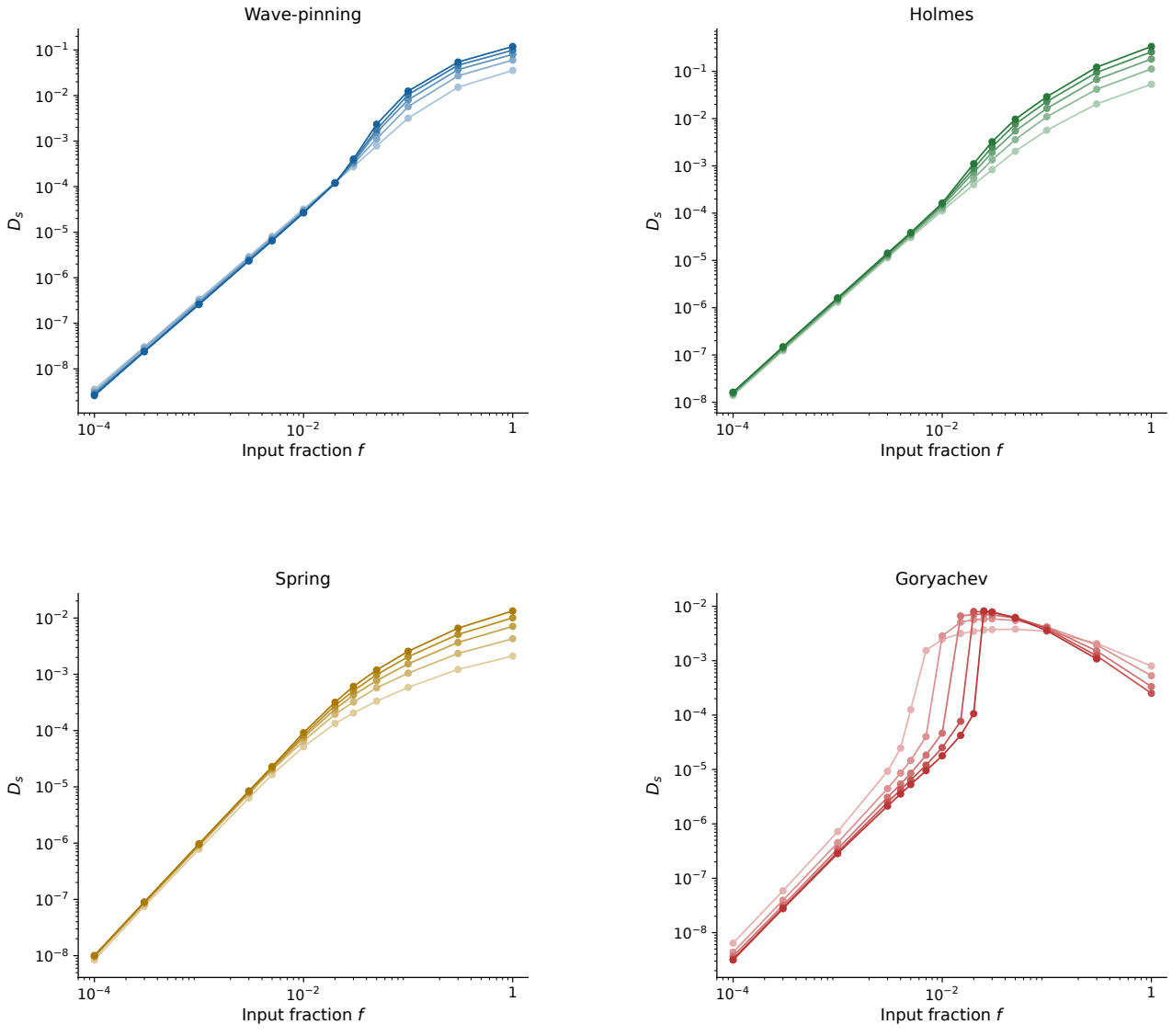


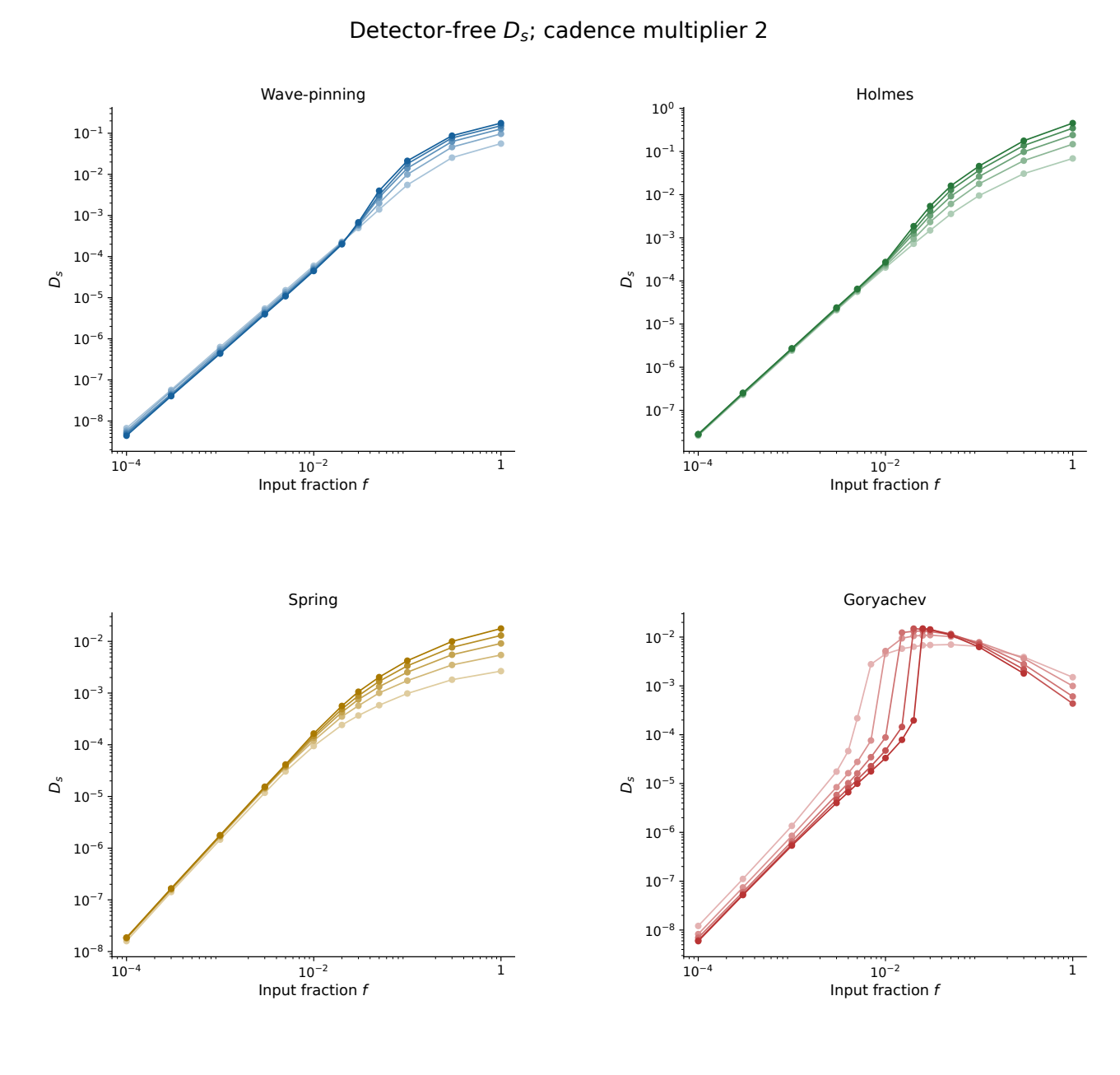


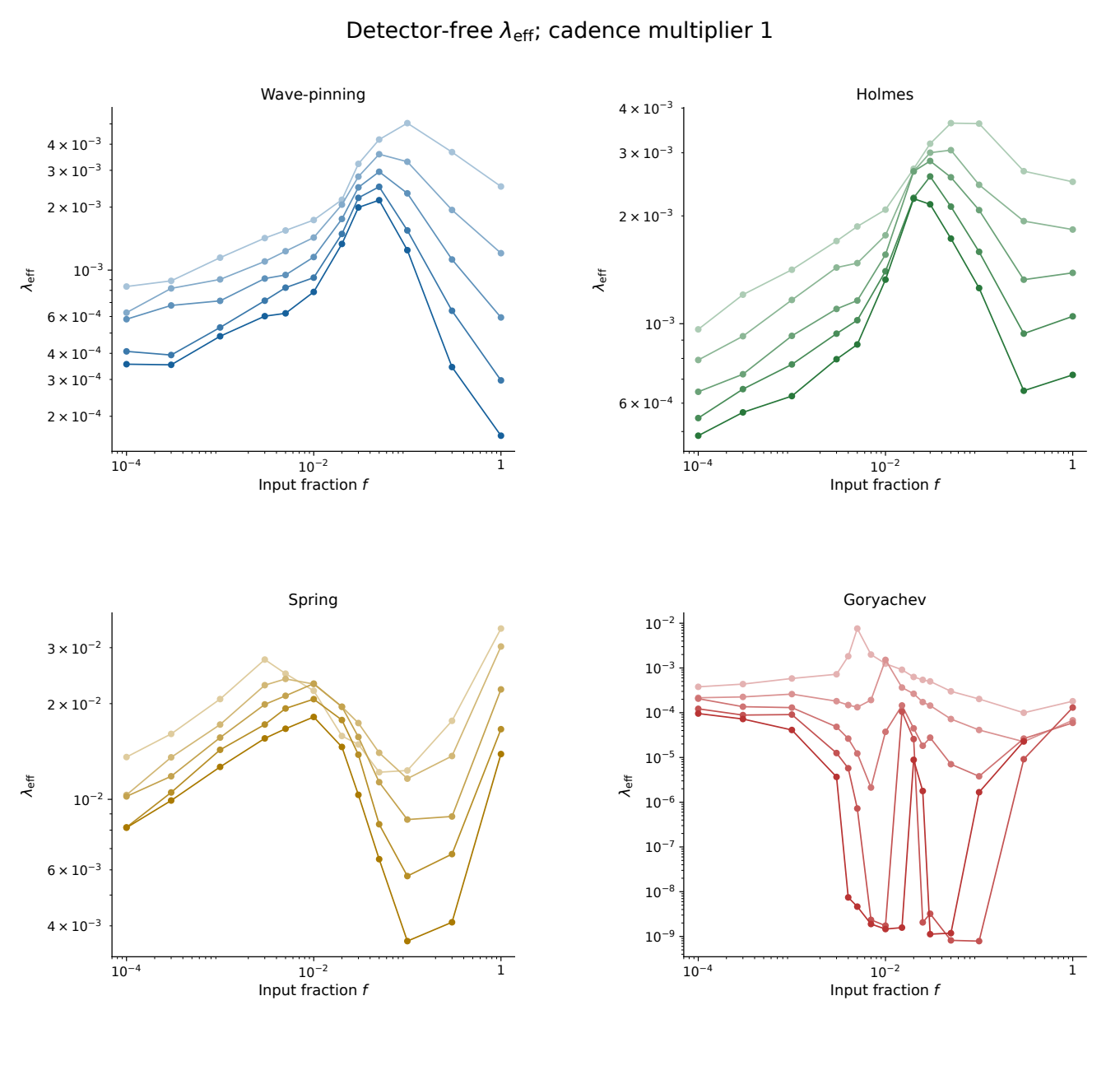


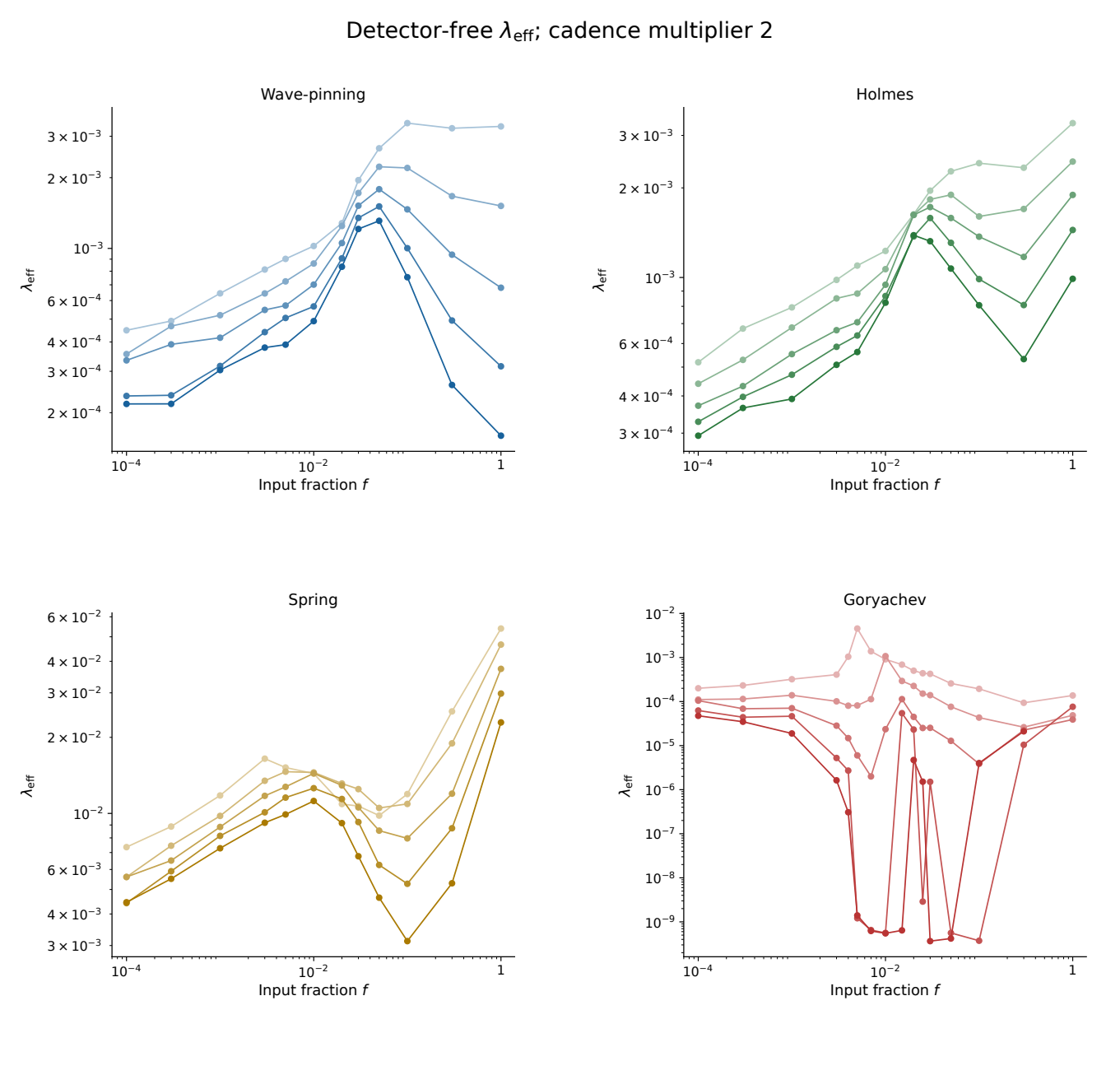


Linear axes retain empirical zeros; only undefined pairs are omitted. Open markers identify forced conditions; no undefined value is imputed.

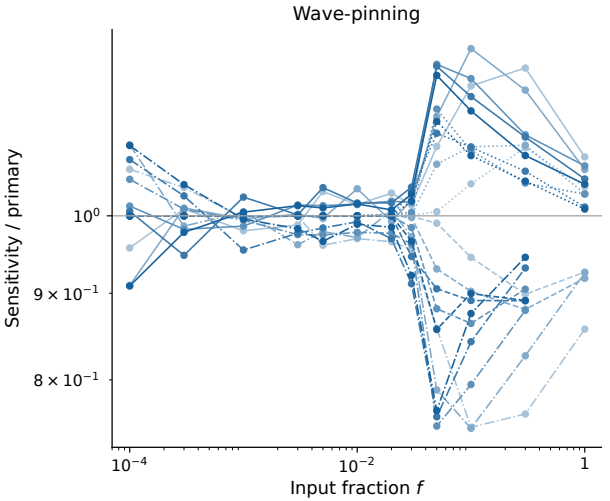

Detector-free D_s; cadence multiplier 1
Wave-pinning
Holmes
Spring
Goryachev
D_s
Input fraction f
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-4}
10^{-2}
1

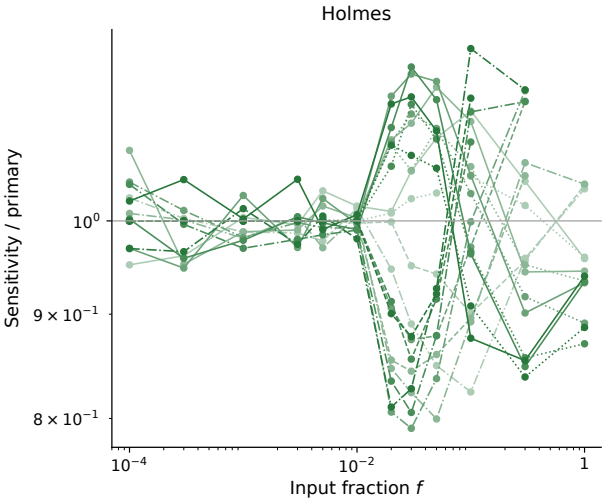

Detector-free D_s; cadence multiplier 2
Wave-pinning
Holmes
Spring
Goryachev
D_s
Input fraction f

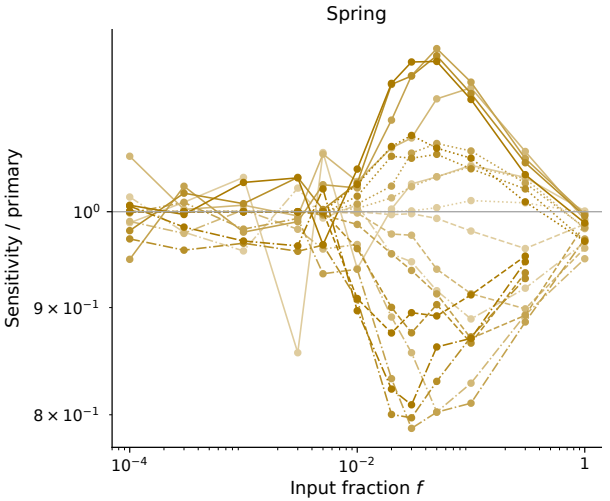

Detector-free λ_{eff}; cadence multiplier 1
Wave-pinning
Holmes
Spring
Goryachev
λ_{eff}
Input fraction f

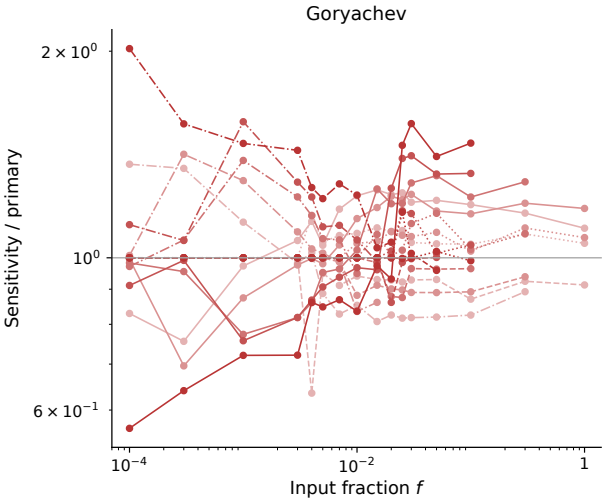

Detector-free λ_{eff}; cadence multiplier 2
Wave-pinning
Holmes
Spring
Goryachev
λ_{eff}
Input fraction f

Tracked sensitivity: D_s

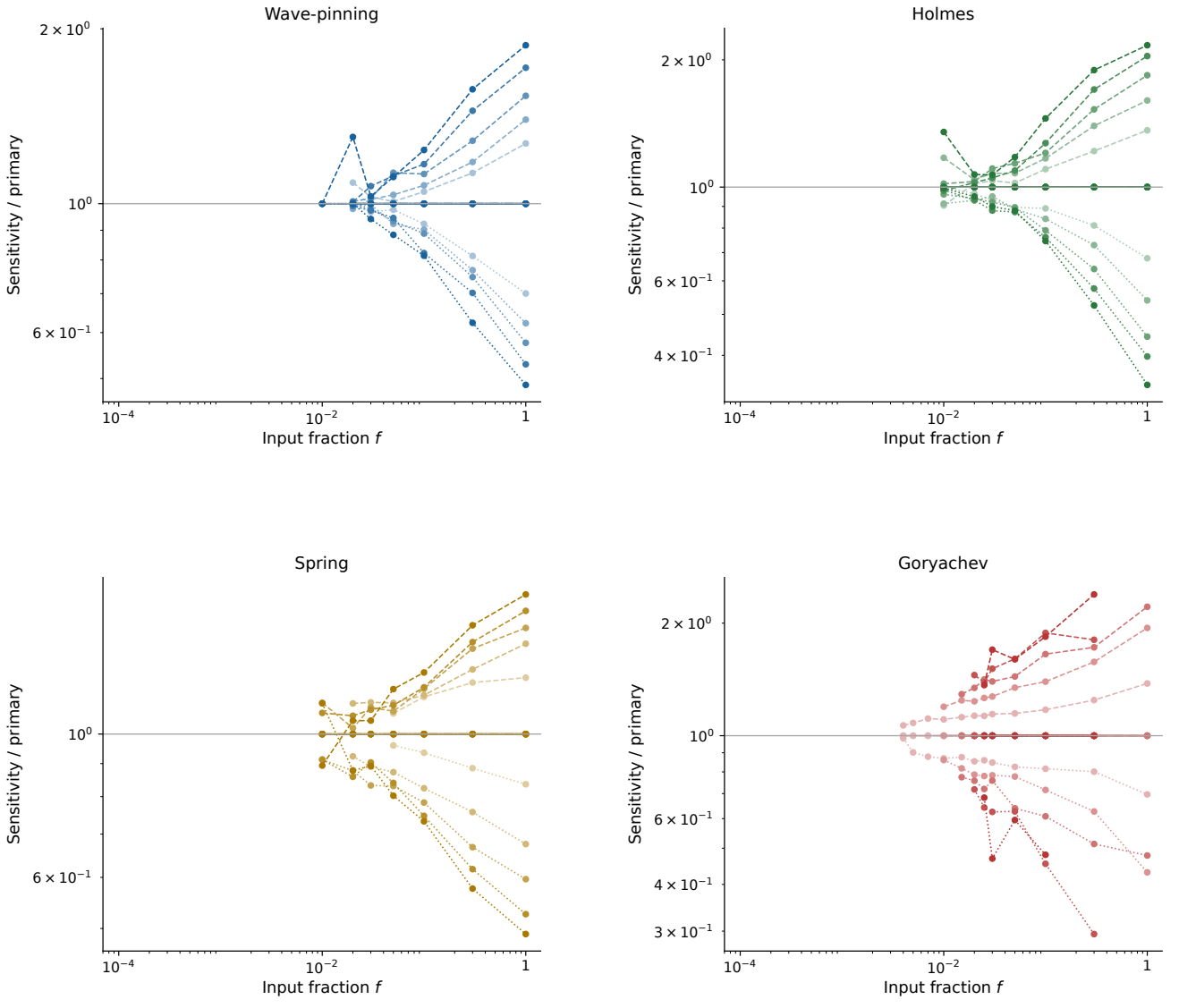




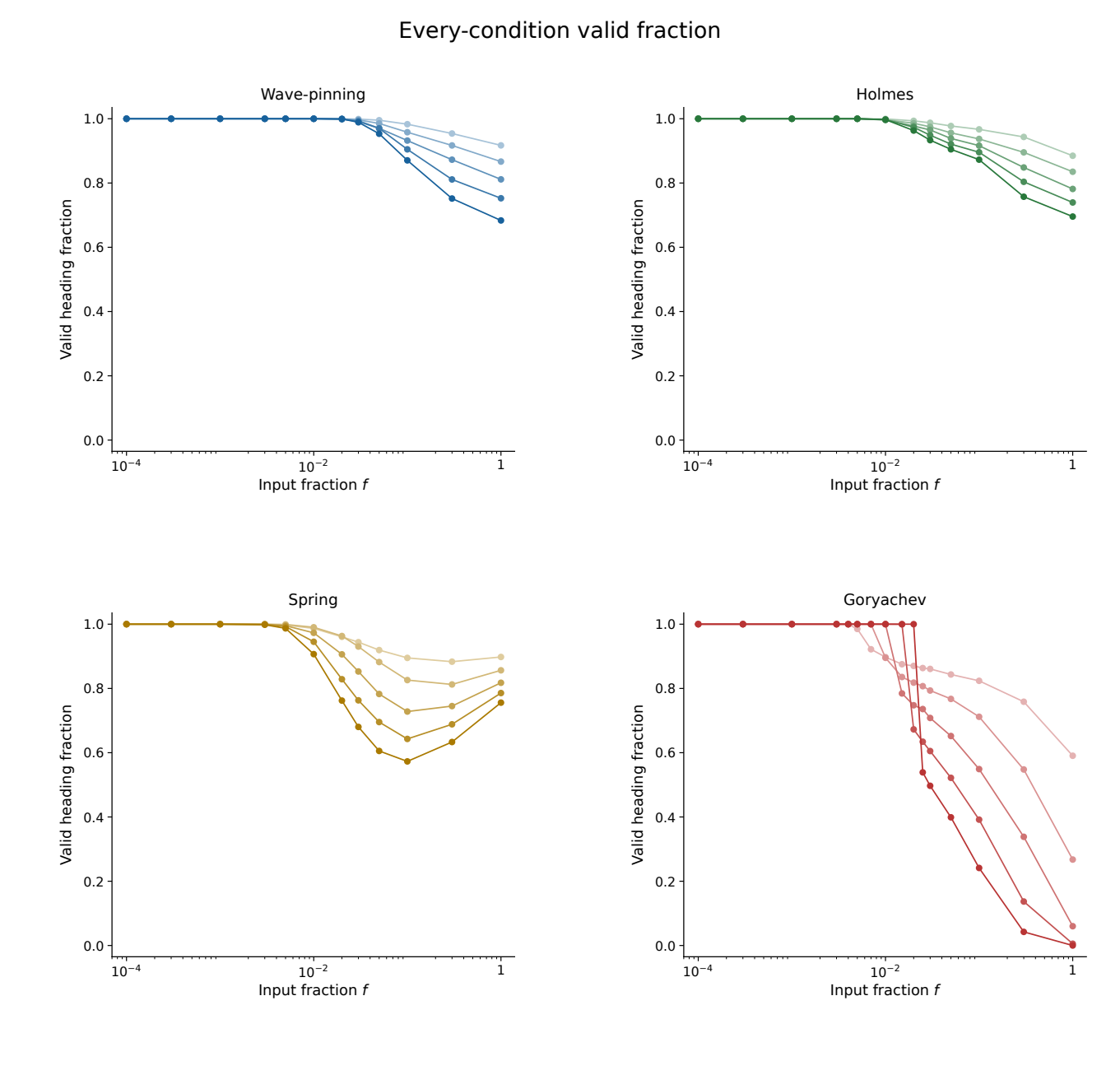


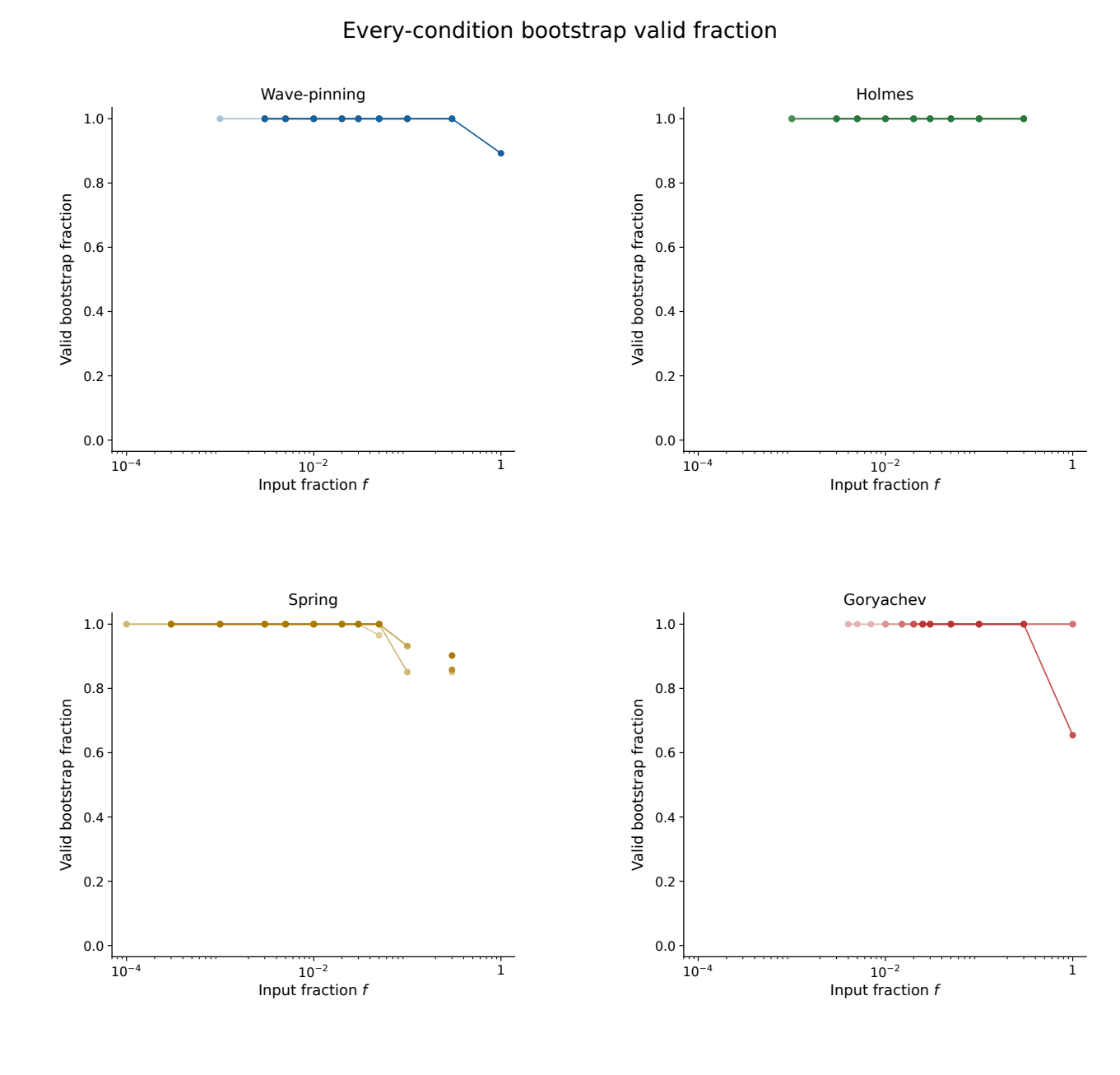


Dashed: hold 2; dotted: hold 10; dash-dot: MSD range 1/8; solid: MSD range 1/2. Primary hold 5, range 1/4.

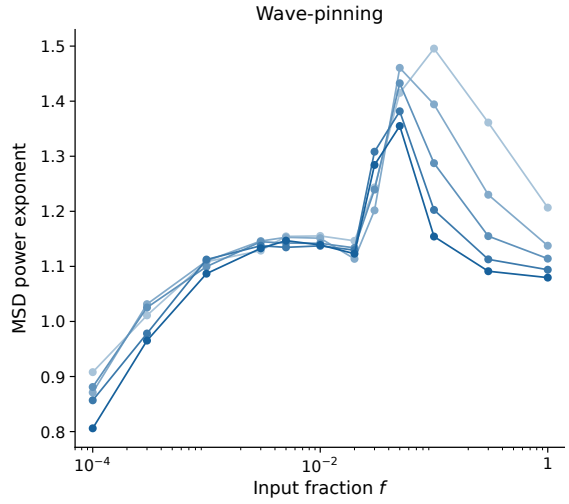


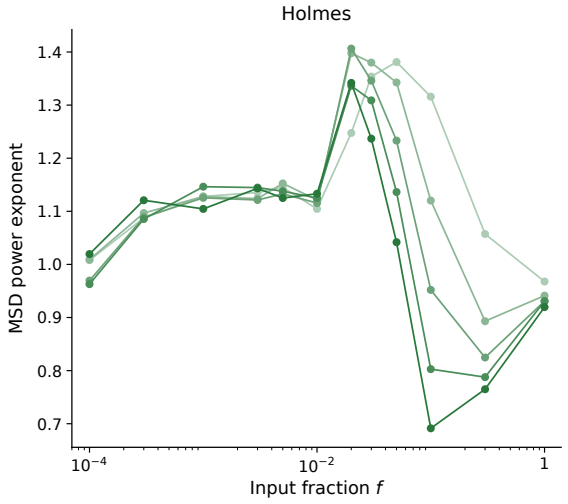
Dashed: hold 2; dotted: hold 10; dash-dot: MSD range 1/8; solid: MSD range 1/2. Primary hold 5, range 1/4.

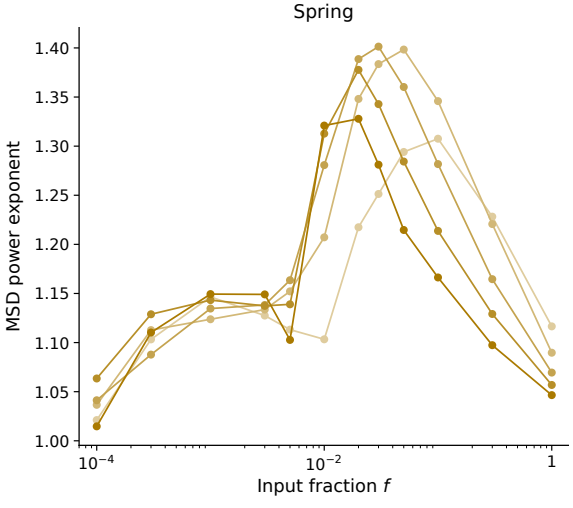

Every-condition valid fraction
Wave-pinning
Holmes
Spring
Goryachev
Valid heading fraction
Input fraction f
0.0
0.2
0.4
0.6
0.8
1.0
10^{-4}
10^{-2}
1

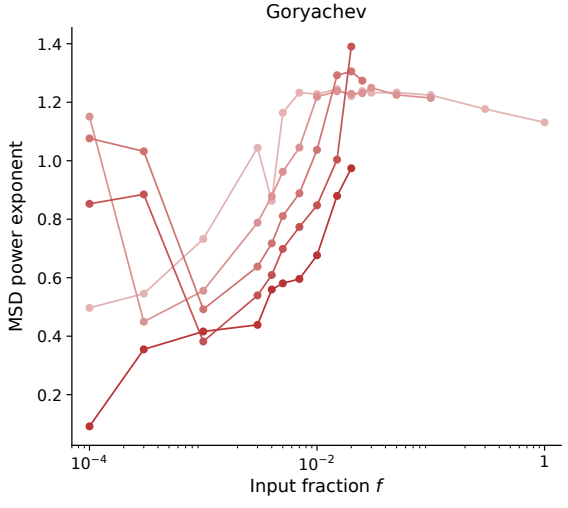

Every-condition bootstrap valid fraction
Wave-pinning
Holmes
Spring
Goryachev
Valid bootstrap fraction
Input fraction f
1.0
0.8
0.6
0.4
0.2
0.0
10^{-4}
10^{-2}
1

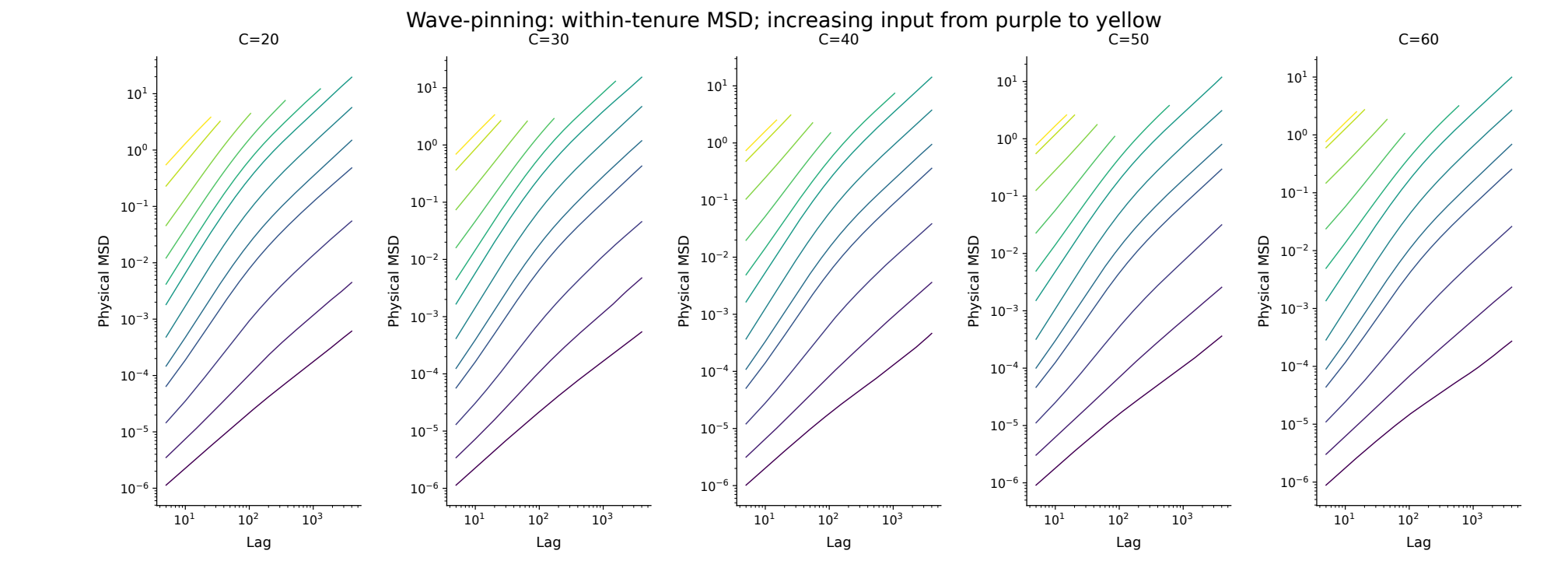
Within-tenure free MSD power exponent

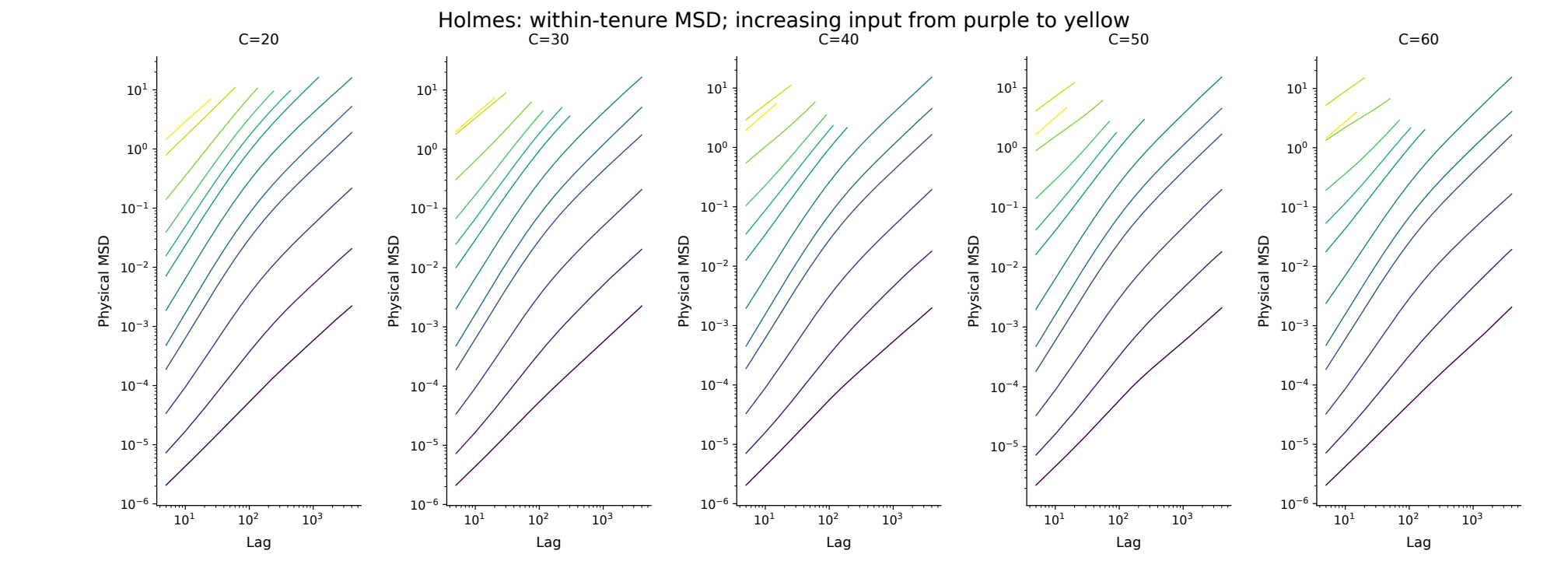


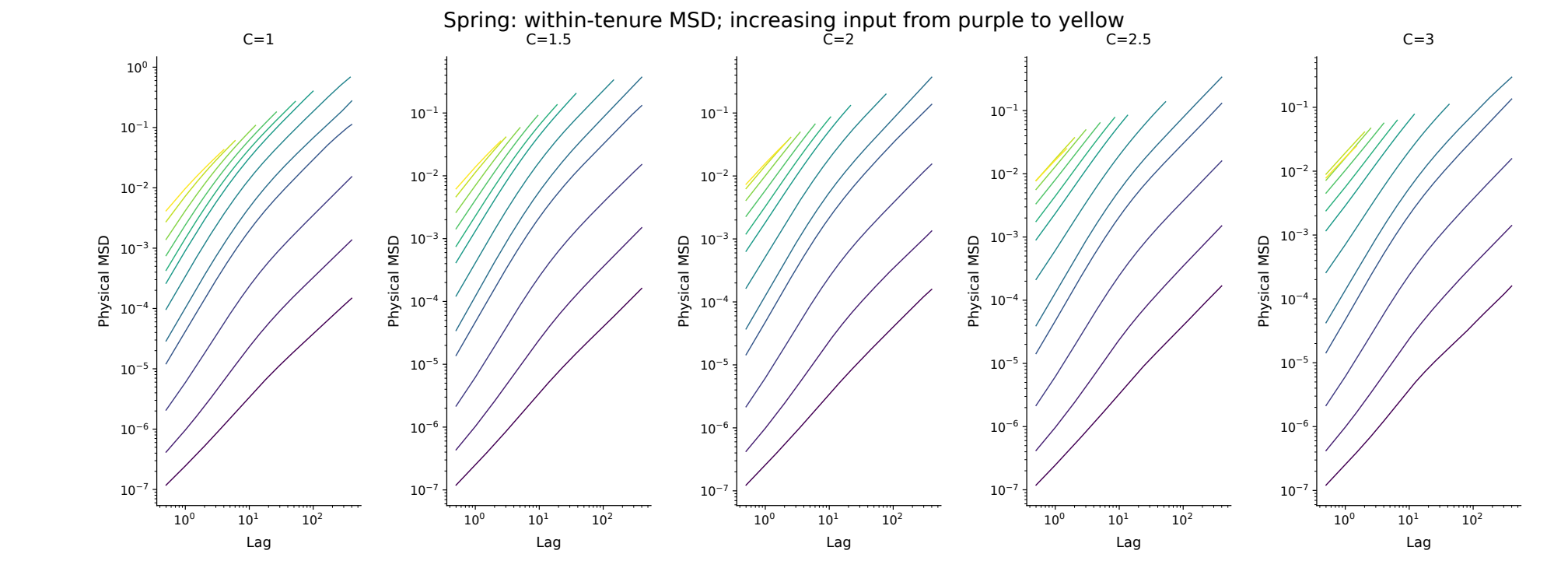


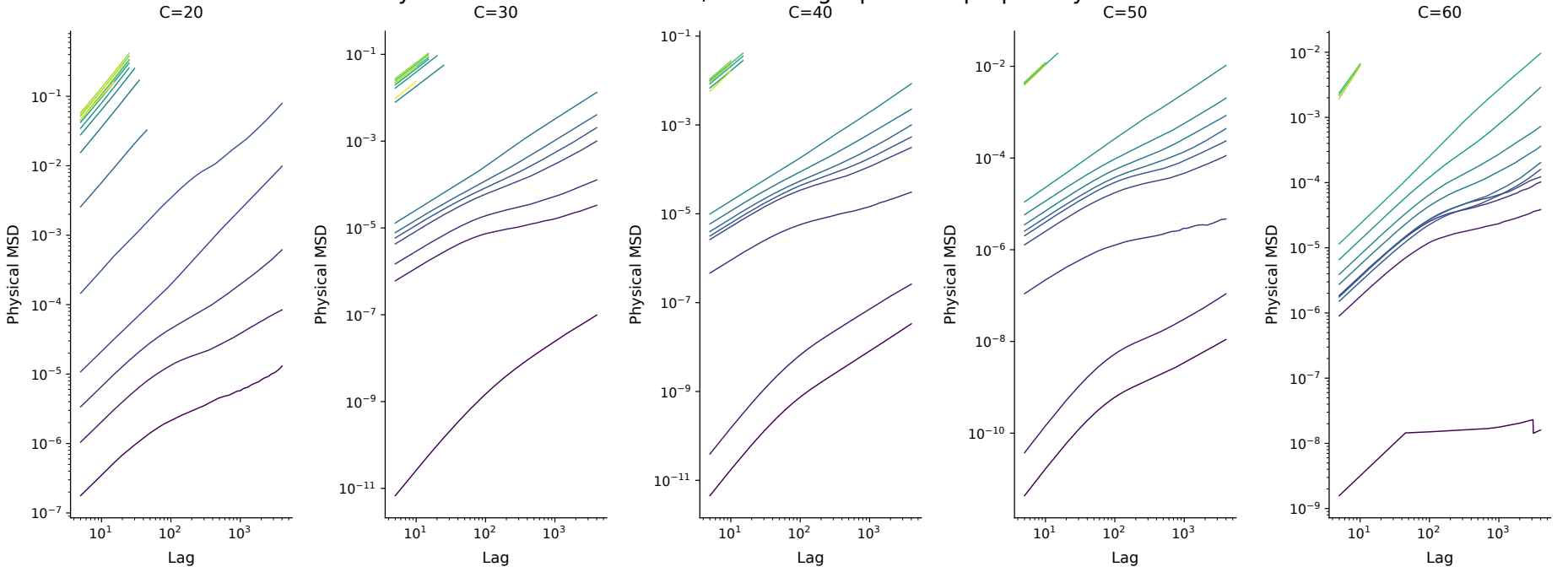


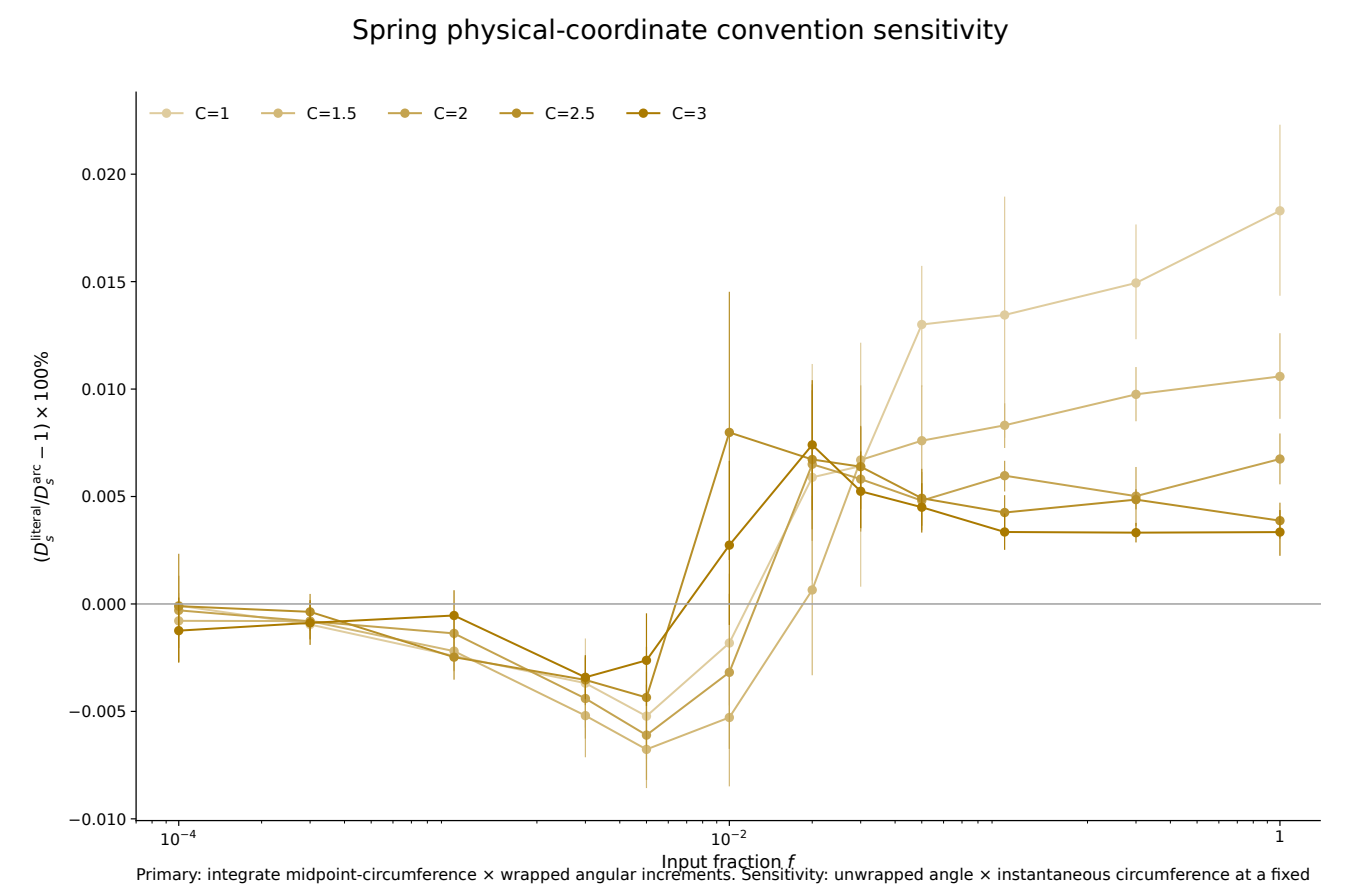



Wave-pinning: within-tenure MSD; increasing input from purple to yellow
C=20
C=30
C=40
C=50
C=60
Physical MSD
Lag


Holmes: within-tenure MSD; increasing input from purple to yellow
C=20
C=30
C=40
C=50
C=60
Physical MSD
Lag

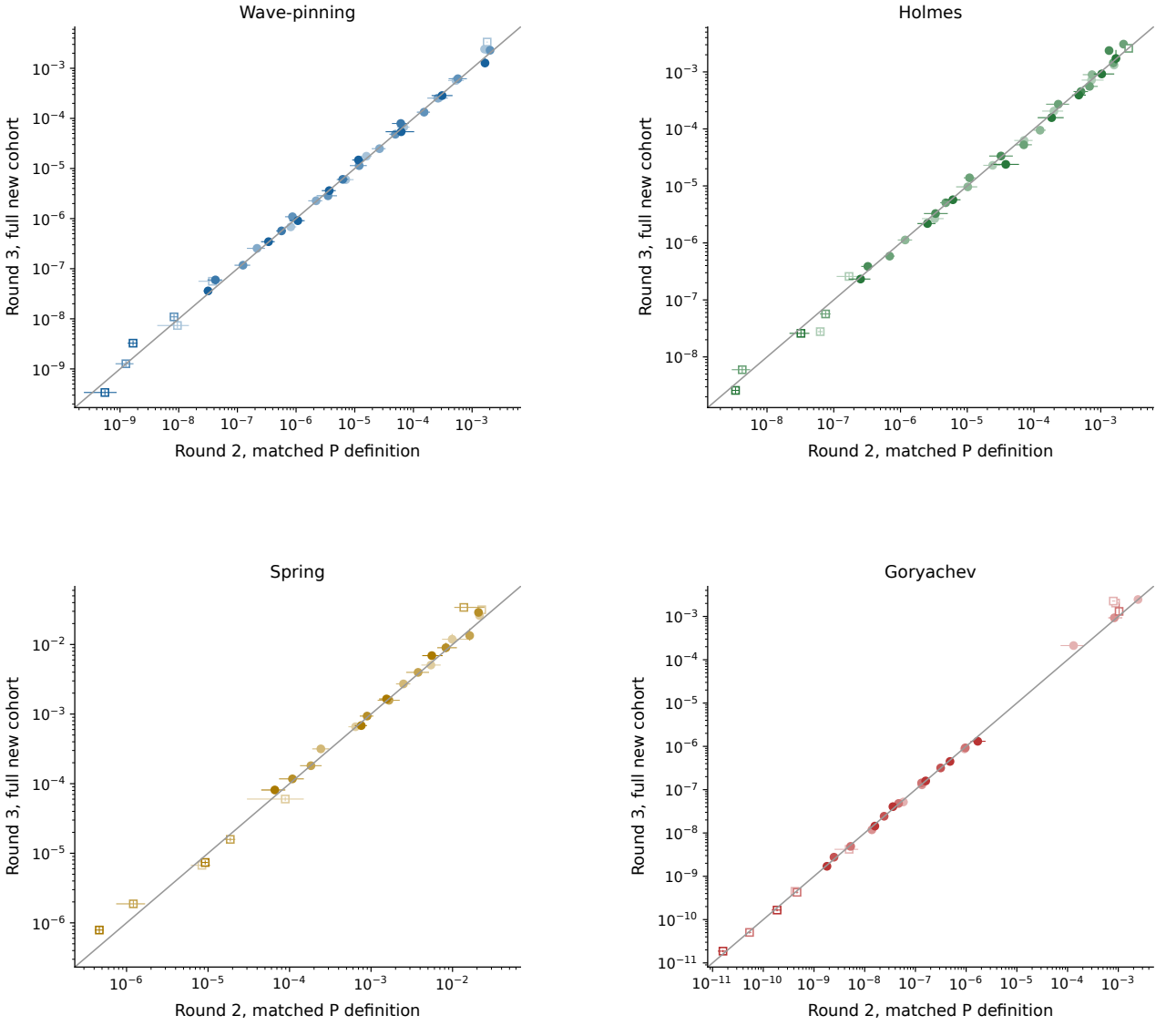

Spring: within-tenure MSD; increasing input from purple to yellow
C=1
C=1.5
C=2
C=2.5
C=3
Physical MSD
Lag


Goryachev: within-tenure MSD; increasing input from purple to yellow
C=20
C=30
C=40
C=50
C=60
Physical MSD
Lag



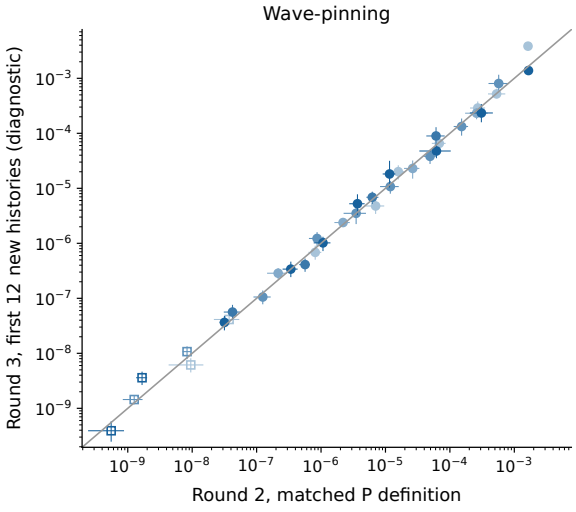
Primary: integrate midpoint-circumference × wrapped angular increments. Sensitivity: unwrapped angle × instantaneous circumference at a fixed coordinate origin. Whole-seed 95% intervals.

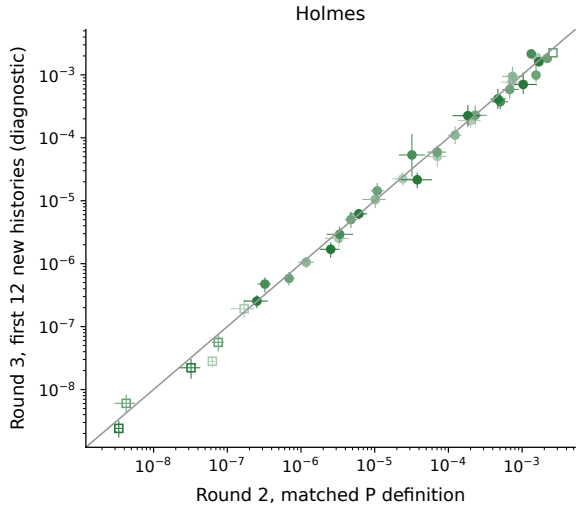
Round-2 versus round-3: Fixed-lag decay rate; full new cohort

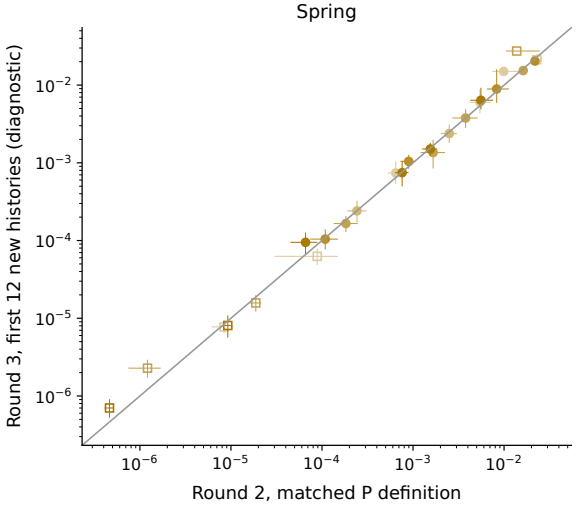


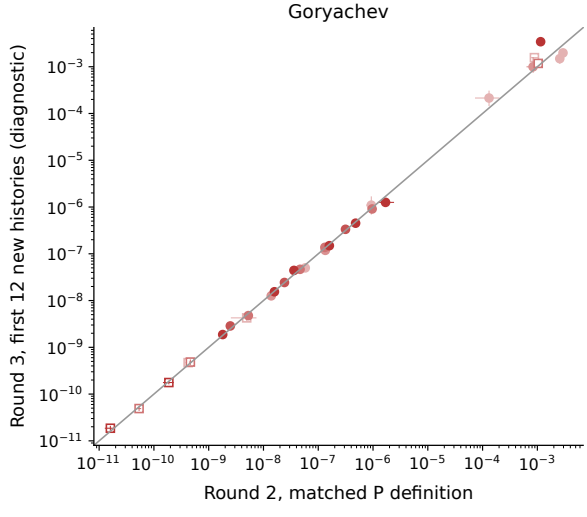
Filled circles: old ensemble has 12 histories. Open squares: old count differs or varies over the tolerance grid. Actual old/new counts and all intervals remain in CSV.

Round-2 versus round-3: Fixed-lag decay rate; first 12 new histories (diagnostic)



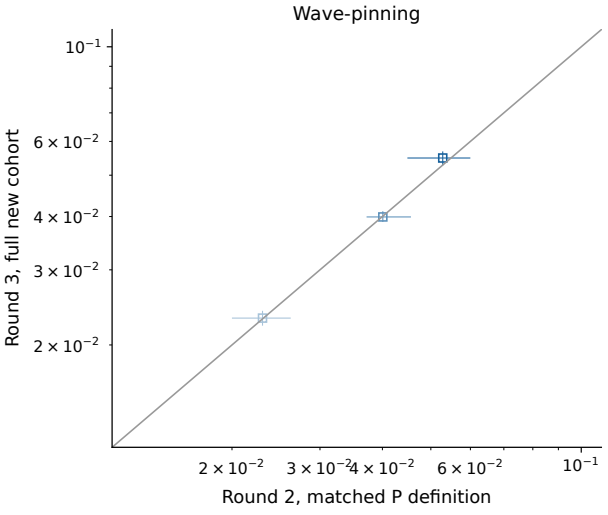


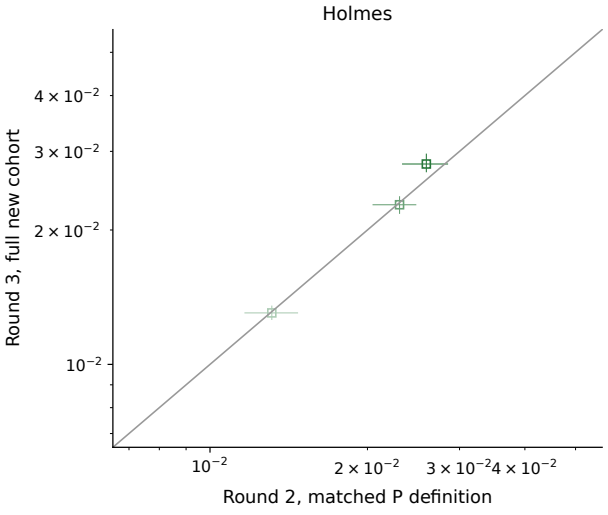


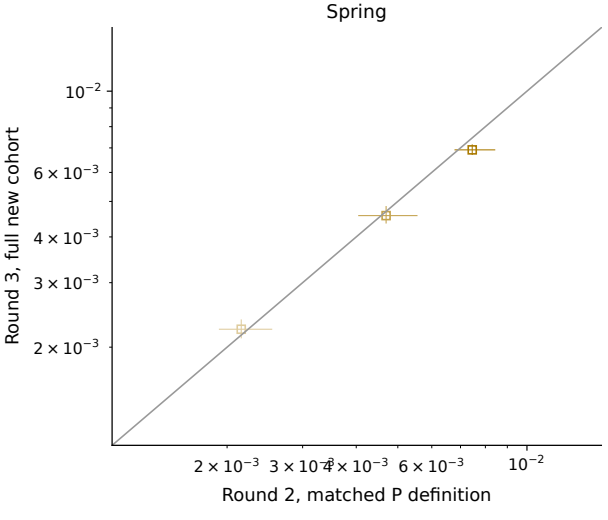


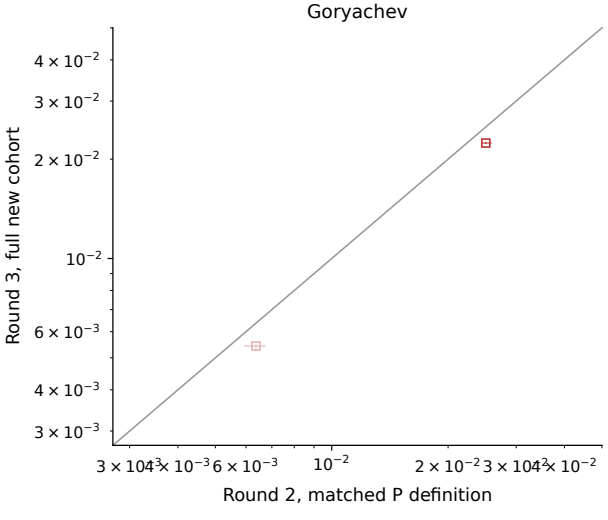
Filled circles: old ensemble has 12 histories. Open squares: old count differs or varies over the tolerance grid. Actual old/new counts and all intervals remain in CSV.

Round-2 versus round-3: Noise tolerance; full new cohort

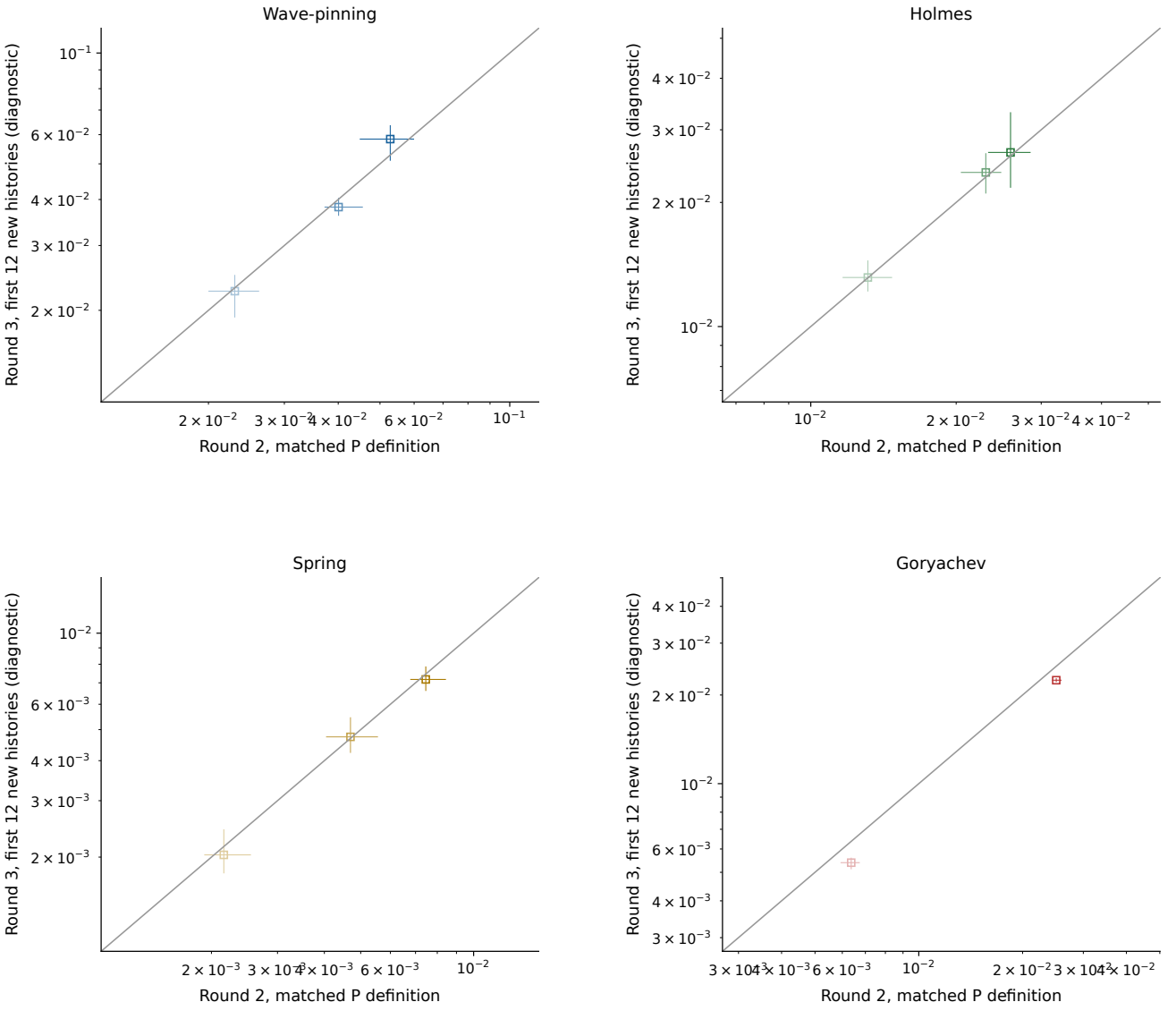








Filled circles: old ensemble has 12 histories. Open squares: old count differs or varies over the tolerance grid. Actual old/new counts and all intervals remain in CSV.

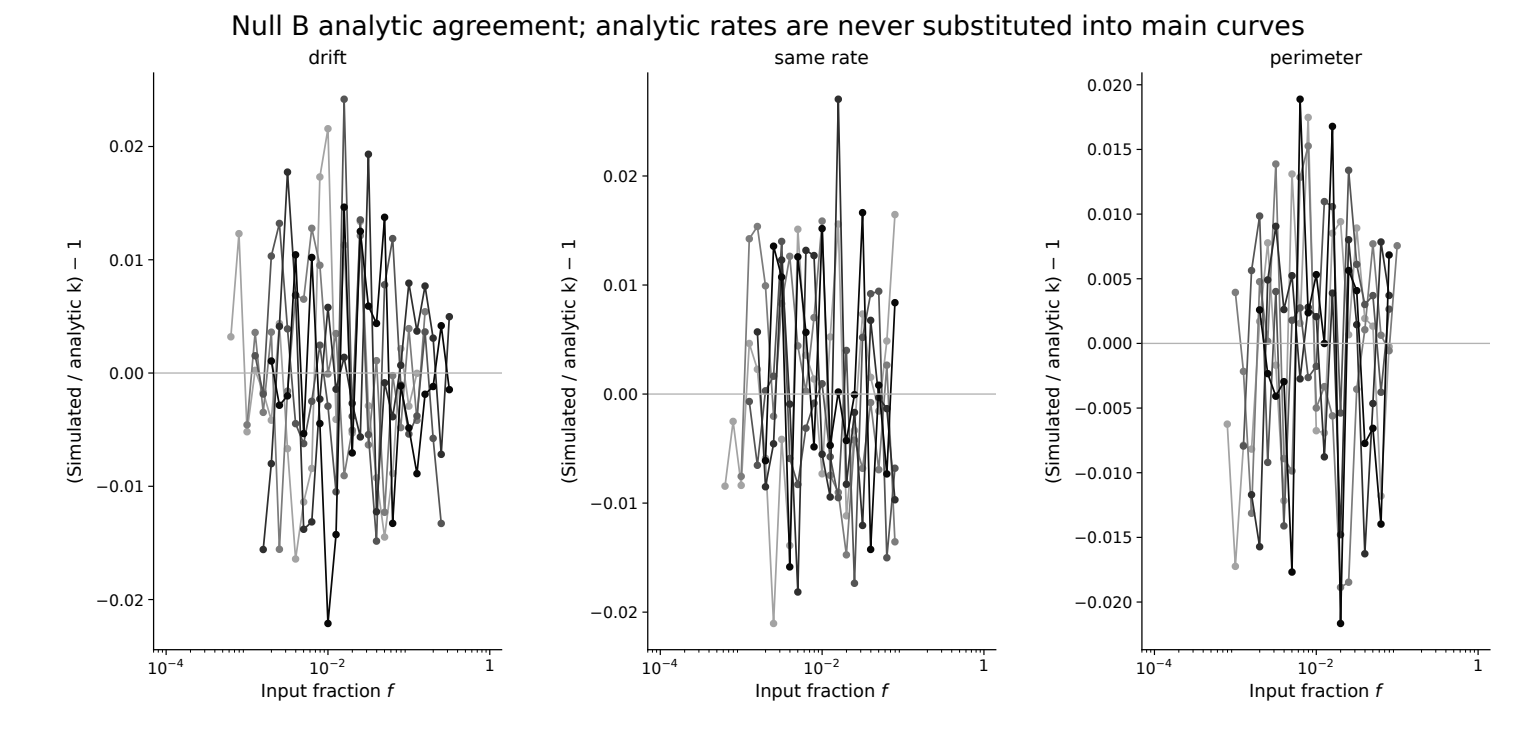

Round-2 versus round-3: Noise tolerance; first 12 new histories (diagnostic)
Wave-pinning
Holmes
Spring
Goryachev
Round 3, first 12 new histories (diagnostic)
Round 2, matched P definition
Filled circles: old ensemble has 12 histories. Open squares: old count differs or varies over the tolerance grid. Actual old/new counts and all intervals remain in CSV.

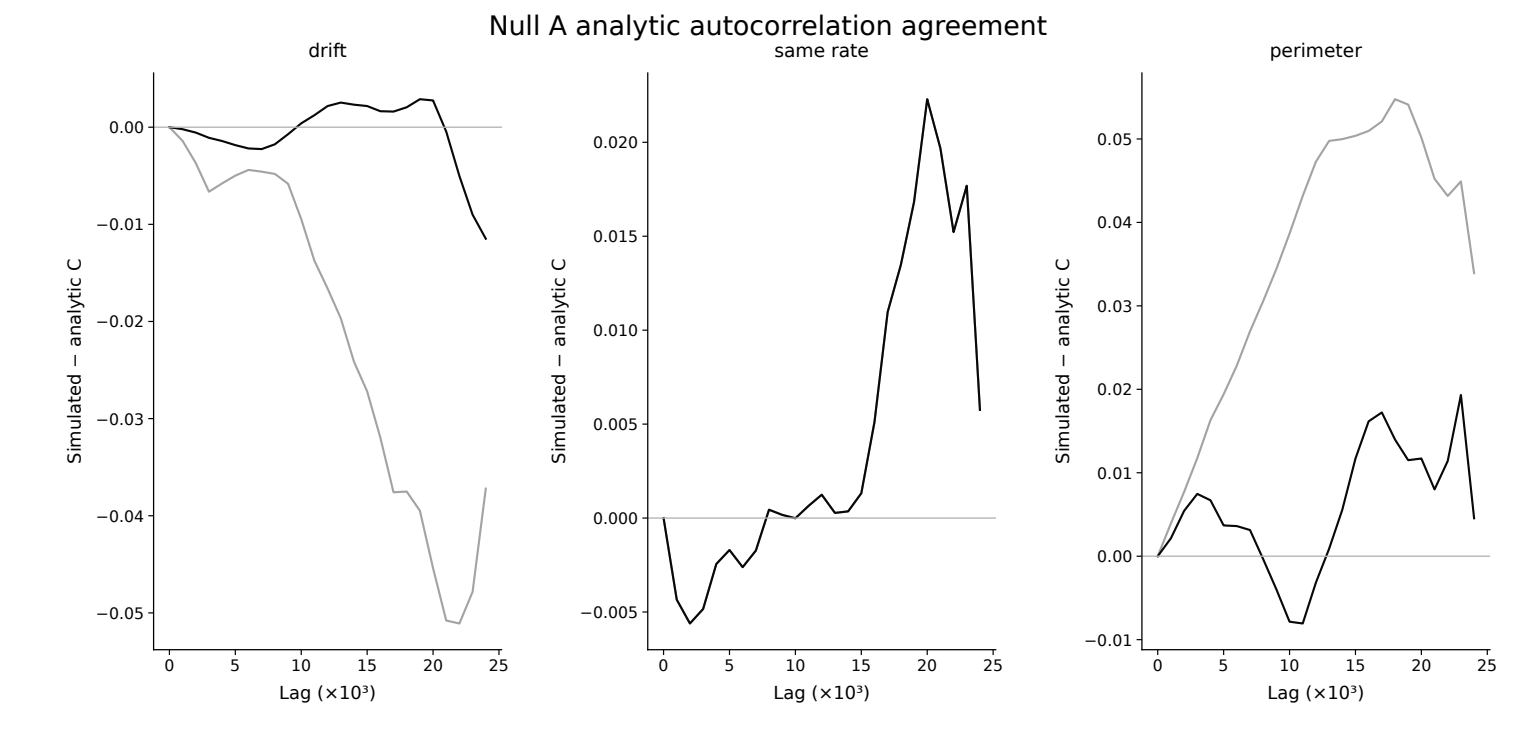
wave_pinning, C=20.0, f=0.001: n=384, horizon=32000.0
wave_pinning, C=20.0, f=0.003: n=384, horizon=32000.0
wave_pinning, C=20.0, f=0.005: n=384, horizon=32000.0
wave_pinning, C=20.0, f=0.01: n=384, horizon=32000.0
wave_pinning, C=20.0, f=0.3: n=384, horizon=32000.0
wave_pinning, C=20.0, f=1.0: n=192, horizon=32000.0
wave_pinning, C=30.0, f=0.003: n=384, horizon=32000.0
wave_pinning, C=30.0, f=0.005: n=384, horizon=32000.0
wave_pinning, C=30.0, f=0.01: n=384, horizon=32000.0
wave_pinning, C=30.0, f=0.02: n=288, horizon=32000.0
wave_pinning, C=30.0, f=0.3: n=192, horizon=32000.0
wave_pinning, C=40.0, f=0.003: n=384, horizon=32000.0
wave_pinning, C=40.0, f=0.005: n=384, horizon=32000.0
wave_pinning, C=40.0, f=0.01: n=384, horizon=32000.0
wave_pinning, C=40.0, f=0.02: n=384, horizon=32000.0
wave_pinning, C=40.0, f=0.03: n=192, horizon=32000.0
wave_pinning, C=50.0, f=0.003: n=384, horizon=32000.0
wave_pinning, C=50.0, f=0.005: n=384, horizon=32000.0
wave_pinning, C=50.0, f=0.01: n=384, horizon=32000.0
wave_pinning, C=50.0, f=0.02: n=384, horizon=32000.0
wave_pinning, C=50.0, f=0.03: n=288, horizon=32000.0
wave_pinning, C=60.0, f=0.003: n=384, horizon=32000.0
wave_pinning, C=60.0, f=0.005: n=384, horizon=32000.0
wave_pinning, C=60.0, f=0.01: n=384, horizon=32000.0
wave_pinning, C=60.0, f=0.02: n=384, horizon=32000.0
wave_pinning, C=60.0, f=0.03: n=384, horizon=32000.0
wave_pinning, C=60.0, f=0.05: n=192, horizon=32000.0
wave_pinning, C=60.0, f=1.0: n=384, horizon=32000.0
holmes_model3, C=20.0, f=0.001: n=384, horizon=32000.0
holmes_model3, C=20.0, f=0.003: n=384, horizon=32000.0
holmes_model3, C=20.0, f=0.005: n=384, horizon=32000.0
holmes_model3, C=20.0, f=0.3: n=192, horizon=32000.0
holmes_model3, C=20.0, f=1.0: n=192, horizon=32000.0
holmes_model3, C=30.0, f=0.001: n=384, horizon=32000.0
holmes_model3, C=30.0, f=0.003: n=384, horizon=32000.0

holmes_model3, C=30.0, f=0.005: n=384, horizon=32000.0
holmes_model3, C=30.0, f=0.01: n=288, horizon=32000.0
holmes_model3, C=30.0, f=1.0: n=384, horizon=32000.0
holmes_model3, C=40.0, f=0.001: n=384, horizon=32000.0
holmes_model3, C=40.0, f=0.003: n=384, horizon=32000.0
holmes_model3, C=40.0, f=0.005: n=384, horizon=32000.0
holmes_model3, C=40.0, f=0.01: n=384, horizon=32000.0
holmes_model3, C=50.0, f=0.001: n=384, horizon=32000.0
holmes_model3, C=50.0, f=0.003: n=384, horizon=32000.0
holmes_model3, C=50.0, f=0.005: n=384, horizon=32000.0
holmes_model3, C=50.0, f=0.01: n=384, horizon=32000.0
holmes_model3, C=50.0, f=0.02: n=192, horizon=32000.0
holmes_model3, C=50.0, f=0.3: n=192, horizon=32000.0
holmes_model3, C=60.0, f=0.003: n=384, horizon=32000.0
holmes_model3, C=60.0, f=0.005: n=384, horizon=32000.0
holmes_model3, C=60.0, f=0.01: n=384, horizon=32000.0
holmes_model3, C=60.0, f=0.02: n=192, horizon=32000.0
spring_ring, C=1.0, f=0.0001: n=384, horizon=3200.0
spring_ring, C=1.0, f=0.0003: n=384, horizon=3200.0
spring_ring, C=1.0, f=0.001: n=384, horizon=3200.0
spring_ring, C=1.0, f=0.05: n=384, horizon=3200.0
spring_ring, C=1.0, f=0.3: n=384, horizon=3200.0
spring_ring, C=1.5, f=0.0001: n=384, horizon=3200.0
spring_ring, C=1.5, f=0.0003: n=384, horizon=3200.0
spring_ring, C=1.5, f=0.001: n=384, horizon=3200.0
spring_ring, C=1.5, f=0.1: n=384, horizon=3200.0
spring_ring, C=1.5, f=0.3: n=192, horizon=3200.0
spring_ring, C=2.0, f=0.0003: n=384, horizon=3200.0
spring_ring, C=2.0, f=0.001: n=384, horizon=3200.0
spring_ring, C=2.0, f=0.05: n=192, horizon=3200.0
spring_ring, C=2.0, f=0.1: n=384, horizon=3200.0
spring_ring, C=2.5, f=0.0003: n=384, horizon=3200.0
spring_ring, C=2.5, f=0.001: n=384, horizon=3200.0
spring_ring, C=2.5, f=0.003: n=288, horizon=3200.0
spring_ring, C=2.5, f=0.1: n=192, horizon=3200.0

Extended horizons and conditions above 96 seeds (71–84)

spring_ring, C=2.5, f=0.3: n=384, horizon=3200.0
spring_ring, C=3.0, f=0.0003: n=384, horizon=3200.0
spring_ring, C=3.0, f=0.001: n=384, horizon=3200.0
spring_ring, C=3.0, f=0.003: n=384, horizon=3200.0
spring_ring, C=3.0, f=0.3: n=384, horizon=3200.0
goryachev, C=20.0, f=0.004: n=384, horizon=32000.0
goryachev, C=20.0, f=0.005: n=192, horizon=32000.0
goryachev, C=30.0, f=1.0: n=192, horizon=32000.0
goryachev, C=40.0, f=1.0: n=384, horizon=32000.0
goryachev, C=50.0, f=0.015: n=192, horizon=32000.0
goryachev, C=50.0, f=0.3: n=384, horizon=32000.0
goryachev, C=50.0, f=1.0: n=384, horizon=32000.0
goryachev, C=60.0, f=0.1: n=192, horizon=32000.0
goryachev, C=60.0, f=0.3: n=384, horizon=32000.0


Null B analytic agreement; analytic rates are never substituted into main curves
drift
same rate
perimeter
(Simulated / analytic k) − 1
Input fraction f
0.02
0.01
0.00
−0.01
−0.02
0.015
0.010
0.005
−0.005
−0.010
−0.015
−0.020
10^{-4}
10^{-2}
1


Null A analytic autocorrelation agreement
drift
same rate
perimeter
Simulated − analytic C
Lag (×10³)
0.00
−0.01
−0.02
−0.03
−0.04
−0.05
0.020
0.015
0.010
0.005
0.000
−0.005
0.05
0.04
0.03
0.02
0.01
0.00
−0.01
0
5
10
15
20
25

Every local neighborhood: circle resolved, cross unavailable; reasons in compact CSV

